

An Introduction to the Subprime Mortgage Sector

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- ▶ **Subprime Mortgage Market:** The subprime mortgage sector has grown rapidly over the past decade and has become a significant sub-sector of the U.S. residential mortgage market. Subprime originations totaled \$613 billion in 2006, 21% of total U.S. residential mortgage originations. As of year-end 2006, the total outstanding balance of subprime mortgages grew to \$1.2 trillion, 12.6% of all mortgage debt.
- ▶ **Subprime Lending:** The lending process in subprime differs significantly from conventional lending. Origination channels are broker-dominated with relatively few loans originated through retail channels. The underwriting process is not standardized and underwriting criteria vary widely among issuers. Furthermore, subprime mortgages are “risk priced,” with less credit worthy borrowers being charged higher interest rates. Finally, subprime lenders are more dependent upon the securitization and secondary whole loan markets for liquidity.
- ▶ **Borrower/Collateral Characteristics:** 2/28 hybrid ARMs are the most popular product for subprime borrowers. Relative to conventional borrowers, the average FICO score of subprime mortgagors is almost 100 points lower and the average loan-to-value ratio is much higher. Furthermore, subprime loans are much smaller in size and tend to have prepayment penalties. Subprime lending caters to cash-strapped borrowers who rely on periodic equity-takeout for debt consolidation and is concentrated in California.
- ▶ **Prepayments:** Prepayment rates on subprime collateral are not as sensitive to changes in interest rate as prime collateral due to fewer options for credit-impaired borrowers in a refi environment and continued cashout refinancing and credit curing in a discount environment.
- ▶ **Credit Performance:** Subprime collateral pools experience elevated delinquency, default, and loss rates relative to prime quality mortgages. However, in subprime collateral, delinquencies are not always a precursor to default. “Rolling delinquencies”, where the borrower is cash flowing but remains contractually delinquent, are common.
- ▶ **Subprime Servicing:** Subprime servicers can have a substantial impact on credit performance and often follow special strategies that are not ordinarily utilized by prime servicers. Higher delinquency, foreclosure, and REO rates result in much higher staffing requirements and costs when compared to prime loans. Subprime servicing fees average around 50 bps, compared to 25 bps for prime loans.
- ▶ **Regulatory Environment:** There are no explicit rules governing subprime lending because, as yet, there is no universally accepted definition of who a subprime borrower is. Instead, regulators have targeted a number of questionable and abusive practices under the general umbrella of “predatory lending.” The relationship between predatory lending and subprime lending stems from the belief that predatory practices are especially endemic to the subprime industry.

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I. Introduction

Subprime mortgage lending is the origination of residential mortgage loans to customers with impaired credit histories. Typically, these borrowers have lower credit scores and/or other credit deficiencies that prevent them from qualifying for prime mortgages. To compensate subprime originators for greater default risks and higher servicing costs, subprime mortgagors pay higher interest rates, points and origination fees. Often termed “B&C” lending, subprime lending is very different from other non-conforming sectors such as “alt-A” or “jumbo,” with subprime borrowers exhibiting increased credit risk and reduced prepayment risk. After origination, the majority of subprime loans are securitized and sold in the capital markets as part of the “Home Equity” ABS sector. While “Home Equity” is used to classify a large gamut of securities backed by residential mortgage loans from prime high LTV mortgages to HELOCs, the vast majority of this sector consists of securitized subprime mortgages.

In this article, we take a detailed look at the subprime mortgage market. We begin by chronicling the rapid growth of the industry and discuss the impetus behind the expansion. Next, we describe the lending process, detailing common origination channels, underwriting procedures and the application of “risk-based” pricing. We document the collateral characteristics of subprime loans and contrast differences among key subprime lenders. We illustrate the reduced prepayment risk and increased credit risk of subprime loans compared to other mortgage products. We take a close look at servicing issues including the importance and prevalence of loss mitigation techniques. Finally, we take a look at the regulatory issues facing the industry. A concluding Appendix provides an in—depth discussion of loan modifications, a loss mitigation technique that has recently attracted a substantial amount of interest.

II. The Subprime Lending Sector

The Explosive Growth of Subprime Lending

The subprime mortgage sector has grown rapidly over the past decade and has become a significant sub-sector of the U.S. residential mortgage market. As shown in Figure 1, subprime mortgage originations reached \$613 billion in 2006, comprising a substantial 21% of all U.S. residential mortgage originations. This is a whopping 411% increase over 2001 origination volume. Similarly, by year-end 2006, total outstanding balance of subprime loans grew to \$1.2 trillion, approximately 12.6% of all outstanding mortgage debt. Concurrently, securitization of subprime loans increased to \$449 billion in 2006, an increase of 416% over 2001. The majority of subprime loans are securitized in the ABS market with 2006 subprime Home Equity ABS issuance totaling 73% of total subprime loan origination volume.

Figure 1: Subprime Issuance Volumes: 2001-2006 (\$bb)

	Loans	MBS
2001	120	87
2002	185	123
2003	310	195
2004	530	363
2005	625	465
2006	613	449

Source: Banc of America Securities

The rapid growth in subprime lending is the result of multiple factors including legislative, economic, and technological developments. Specifically, federal legislation enacted by Congress in the early 1980s pre-empting restrictive state mortgage laws helped formed the legal basis for the modern subprime mortgage industry. Specifically, the new federal regulations permitted lenders to charge high rates and fees to borrowers and allowed variable interest rates and balloon payments on mortgage products. In addition, the Tax Reform Act of 1986 (TRA), which eliminated interest rate deductions on non-mortgage consumer debt, increased the financial attractiveness of mortgage debt, as it became an economic way to consolidate other forms of consumer debt. Most importantly, robust home price appreciation and growing investor acceptance of MBS and ABS provided the necessary equity and liquidity to fuel and fund this rapidly growing market. Lastly, technological developments such as automated underwriting and credit scoring greatly increased originator capacity and their ability to more accurately quantify credit risk.

The Lending Process

The lending process in subprime differs significantly from conventional lending in many ways. Subprime lending is a broker-dominated business with comparatively few loans originated through retail channels. Also, the underwriting process is not standardized and underwriting criteria vary widely among issuers. Subprime mortgages are “risk priced,” with less credit worthy borrowers being charged much higher interest rates. Finally, subprime lenders are more dependent upon the securitization and secondary whole loan markets for liquidity.

In general, mortgages are originated through one of three channels: broker, correspondent, or retail. Figure 2 shows the percentage of subprime mortgages originated through each channel during 2006. Mortgage brokers, the most common origination channel, act as

agents and simply bring together originators with borrowers. They do not control the underwriting process, nor do they have any source of liquidity for funding the origination. For their service, brokers command a fee that is passed on to the borrower. Brokers often work with more than one lender, and large lenders can work with thousands of brokers. In contrast, correspondents are responsible for both the underwriting process and the initial funding of the mortgage. Correspondents draw on warehouse lines from financial institutions for liquidity. The loans are then sold in the secondary markets and may be re-underwritten by the buyer. Collectively, correspondent and broker loans are considered “wholesale” originations, as the ultimate originator did not have direct contact with the borrower. In the retail channel, the originator is directly involved with the applicant in the underwriting process via telephone, the Internet, and/or traditional “brick and mortar” outlets. All other things being equal, retail originations are considered to be less risky than wholesale originations, as the originator was directly involved with the applicant during the underwriting process.

Figure 2: 2006 Subprime Origination Channel Distribution

Broker	Correspondent	Retail
61	21	18

Source: Banc of America Securities Estimates based on National Mortgage News Data

Underwriting in the mortgage market has evolved over the years and credit is extended by using a combination of risk-based pricing (price rationing) and risk-based qualification (non-price rationing). On one end of the spectrum is the prime conventional mortgage market where underwriting guidelines are effectively controlled by the GSEs. Proprietary automated underwriting systems developed by the GSEs are used by traditional prime lenders to qualify applicants based on agency underwriting guidelines. Mortgage rates for approved applicants are similar among lenders and fall within a narrow band across credit grades around the “average” rate. On the other end of the spectrum is subprime mortgage lending where no such standardized underwriting guidelines exist. Each subprime lender defines a unique underwriting matrix to qualify borrowers and set mortgage rates.

Requirements to qualify an application for a subprime mortgage are not as stringent as for a prime mortgage. Approved applicants are charged a rate depending upon the level of perceived risk which depends on a number of factors and is specified in the underwriting matrix. Therefore, mortgage rates vary significantly across borrowers and there is no generic “subprime rate” lenders can quote.

To develop an understanding of the mechanics of risk-based pricing in the subprime underwriting process, let’s walk through the details of a specific mortgage program. We focus on Option One’s “Latitude Advantage” program. Under this program, a borrower is qualified and the mortgage rate is determined based on various factors which include the applicant’s credit score, risk grade, income documentation, LTV, and use and type of property. The risk grade for an applicant under the “Latitude Advantage” program is determined primarily based on prior mortgage payment history and could be assigned one of six possible values ranging from “AA+” to “CC”. In Figure 3 we summarize criteria for assigning risk grades based on late payment history. Note that the definition of what constitutes an “A” borrower versus a “B” borrower is somewhat arbitrary and varies widely among lenders. An applicant that is classified as a “B” grade in one underwriting matrix might be classified as a “C” grade in another. This can result in a large dispersion in interest rates charged to an applicant by different lenders. Some lenders may place more

emphasis on credit scores or delinquency history, while others will be more restrictive on recent bankruptcy.

Figure 3: Borrower Risk Grades in Subprime

Risk Category	Mortgage Delinquencies within Past 12 Months
AA+	0X30
AA	1X30
A	2X30
B	4X30 or 2X30, 1X60
C	6X30, 1X60, 1X90 or 6X30, 2X60, 0X90
CC	Case by Case Basis

1X30 means one late payment

Source: Option One Deal Prospectus, Banc of America Securities

The mortgage rate matrix for the “Latitude Advantage” program is specified along four primary dimensions: risk grade, credit score, LTV, and documentation. The mortgage rate or limits on these dimensions are further refined based on additional attributes like loan size, property type etc. The two year prepay penalty 2/28 ARM mortgage rate matrix effective on 5/25/2007 for the eastern region is shown in Figure 4. The “risk pricing” of mortgages is evident from mortgages with higher credit risk requiring higher mortgage rates. However, there are usually clear limits within which a loan may be underwritten regardless of mortgage rates (risk-based qualification). In the “Latitude Advantage” program, the constraint on LTV is used as the primary lever to control the risk associated with the loan. All the other dimensions are allowed to vary in relatively broad ranges. For example, borrowers in the “CC” risk category are allowed only 65% LTV loans. Similarly, borrowers who state their income generally have lower maximum allowed LTVs. The maximum debt-to-income (DTI) ratio allowed under the “Latitude Advantage” is very high at 55% across credit grades compared to an average DTI of 32% in prime agency mortgages. A high backend DTI is a fairly normal aspect of subprime lending. Furthermore, the program guidelines generally require bankruptcies be discharged, dismissed, or paid off prior to funding.

Figure 4: Subprime Mortgage Rates based on Various Borrower/Loan Characteristics

Grade	Credit Score LTV >>	Full Doc						Stated Income/Lite					
		65%	70%	75%	80%	85%	90%	65%	70%	75%	80%	85%	90%
AA+	700+	7.00	7.15	7.30	7.40	8.00	8.40	8.15	8.30	8.45	8.55	9.20	10.00
	680	7.10	7.25	7.40	7.45	8.05	8.55	8.25	8.40	8.55	8.60	9.35	10.15
	660	7.15	7.30	7.35	7.50	8.15	8.55	8.30	8.45	8.50	8.65	9.45	10.25
	640	7.30	7.50	7.55	7.80	8.45	8.85	8.45	8.65	8.80	8.90	9.70	10.50
	620	7.35	7.55	7.60	7.85	8.50	8.90	8.65	8.95	9.00	9.10	10.00	10.80
	600	7.50	7.75	7.95	8.25	8.80	9.20	8.70	8.95	9.25	9.40	10.30	11.20
	580	7.80	8.05	8.30	8.65	9.25	9.65	9.10	9.35	9.70	10.00	10.85	11.65
	560	8.05	8.35	8.55	9.00	9.65	10.10	9.45	9.75	10.15	10.45		
AA	700+	8.75	9.05	9.40	9.90	10.55							
	680	7.35	7.50	7.65	7.75	8.25	8.65	8.45	8.60	8.75	8.85	9.55	10.35
	660	7.45	7.60	7.75	7.80	8.40	8.80	8.55	8.70	8.85	8.90	9.70	10.50
	640	7.50	7.65	7.70	7.85	8.40	8.80	8.60	8.75	8.80	8.95	9.80	10.60
	620	7.65	7.85	7.90	8.00	8.55	8.95	8.75	8.95	9.10	9.20	9.95	10.75
	600	7.70	7.90	7.95	8.05	8.60	9.00	8.95	9.25	9.30	9.40	10.25	11.05
	580	7.75	8.00	8.20	8.35	8.90	9.30	9.10	9.35	9.65	9.80	10.65	11.45
	560	8.05	8.30	8.55	8.75	9.35	9.75	9.40	9.65	10.00	10.30	11.20	12.00
A	700+	8.30	8.60	8.80	9.10	9.75	10.20	9.75	10.05	10.35	10.65		
	680	9.00	9.30	9.65	10.00	10.65							
	660	7.45	7.60	7.75	7.85	8.35	8.75	8.55	8.70	8.85	8.95	9.65	10.55
	640	7.55	7.70	7.85	7.90	8.50	8.90	8.65	8.80	8.95	9.00	9.80	10.70
	620	7.60	7.75	7.80	7.95	8.50	8.90	8.70	8.85	8.90	9.05	9.90	10.70
	600	7.75	7.95	8.00	8.10	8.65	9.05	8.90	9.10	9.15	9.25	10.10	10.90
	580	7.80	8.00	8.05	8.15	8.70	9.10	9.15	9.35	9.40	9.60	10.35	11.15
	560	7.85	8.10	8.30	8.45	9.00	9.40	9.30	9.55	9.75	9.90	10.75	11.55
B	700+	8.15	8.40	8.65	8.85	9.45	9.85	9.70	9.95	10.20	10.40		
	680	8.40	8.70	8.90	9.20	9.85	10.30	9.95	10.25	10.50	10.80		
	660	9.10	9.40	9.75	10.10								
	640	7.85	8.05	8.25	8.40	8.90	9.30	8.95	9.15	9.35	9.50	10.20	11.10
	620	8.05	8.25	8.45	8.55	9.30	9.75	9.10	9.30	9.50	9.60	10.55	11.50
	600	8.10	8.30	8.45	8.60	9.50	10.00	9.30	9.50	9.70	9.85	11.05	11.95
	580	8.30	8.55	8.70	8.75	9.80	10.30	9.50	9.75	9.90	9.95	11.30	12.20
	560	8.40	8.70	8.85	8.90	10.15	10.65	9.75	10.05	10.20	10.30	11.80	12.70
C	700+	8.50	8.80	9.05	9.10	10.50	11.00	9.95	10.25	10.40	10.55	12.25	
	680	8.85	9.10	9.45	9.60	11.10	11.55	10.40	10.65	11.00	11.15		
	660	9.15	9.40	9.70	10.00			10.75	10.95	11.30	11.60		
	640	9.85	10.05	10.60	10.95								
	620	8.20	8.40	8.60	8.75	9.25	9.70	9.30	9.50	9.70	9.85	10.55	11.50
	600	8.35	8.55	8.75	8.85	9.60	10.15	9.45	9.65	9.85	9.95	10.90	11.95
	580	8.45	8.65	8.80	9.00	9.85	10.50	9.50	9.70	9.70	9.90	11.05	12.10
	560	8.65	8.90	9.05	9.10	10.15	10.85	9.75	10.10	10.25	10.30	11.65	12.75
CC	700+	8.75	9.05	9.20	9.25	10.50	11.25	10.00	10.25	10.40	10.55	12.00	13.15
	680	8.85	9.15	9.40	9.50	10.85	11.65	10.30	10.60	10.85	10.95	12.60	
	660	9.30	9.55	9.90	10.10	11.55		10.90	11.15	11.50	11.70		
	640	9.50	9.70	10.05	10.40			11.10	11.30	11.65	12.00		
	620	10.20	10.40	10.95									
	600												
	580												
	560												

Source: Option One Web Site, Banc of America Securities

Further complicating the subprime lending process is the notion of underwriting exceptions as some lenders will underwrite borrowers who fall outside their underwriting matrix. In such cases, the underwriting matrix is only used as a guideline, with the final risk classification based on other external factors. While many lenders grant few exceptions, some lenders grant exceptions in up to 50% of approved loans. However, this is not to say all applicants are approved. In fact, application denial rates can exceed 50% for subprime

lenders, much higher than for prime lenders. Subprime borrowers also pay higher points and origination fees than conventional borrowers, although exact magnitudes are difficult to ascertain.

Finally, the funding sources for subprime originators differ from conventional prime originators. Prime conventional lenders are usually large depository institutions that may hold many loans in their portfolios. In contrast, most subprime originators are specialty lenders without access to deposits for cheap funding. Subprime lenders are typically dependent upon the secondary whole loan and securitization markets for liquidity.

The Lenders

The subprime mortgage industry is dominated by a relatively small number of lenders. In fact, the top 10 originators controlled over 50% of the subprime market in 2006. Figure 5 shows the origination volumes of the largest subprime lenders in 2006. Notice how the preferred origination channels differ among the top lenders. For example, Option One, New Century Financial, Fremont, and First Franklin originate largely through mortgage brokers. In contrast, Washington Mutual and Wells Fargo origination is dominated by the correspondent channel. Interestingly, Countrywide and HSBC production is balanced across all the three origination channels.

Figure 5: Top 10 Subprime Originators in 2006 and their Origination Channels

Originator	Market	Retail	Broker	Correspondent
	Share			
Wells Fargo Home Mortgage	10.3%	25%	9%	66%
HSBC Finance	7.3%	33%	40%	27%
New Century Financial Corp.	7.1%	8%	92%	0%
Countrywide Financial Corp.	5.6%	38%	33%	29%
WMC Mortgage Corp.	4.5%	0%	87%	13%
Fremont Investment & Loan	4.4%	0%	100%	0%
Option One Mortgage Corp.	4.1%	7%	92%	1%
First Franklin Financial	3.8%	8%	91%	1%
Washington Mutual	3.7%	0%	39%	61%
Ameriquest Mortgage Corp.	3.5%	26%	74%	0%

Source: National Mortgage News, Banc of America Securities

Different originators use different methods to fund the loans that they originate. Subsidiaries of well capitalized, highly rated, depository institutions, like HSBC Finance, with access to cheap funding through deposits can simply portfolio their originations. The majority of originators that specialize primarily in subprime lending either securitize the loans that they originate on their own shelves (First Franklin, Ameriquest) or sell their production as whole loans in the secondary market (Option One, New Century) which is then securitized via Wall Street conduits

III. Characteristics of Subprime Collateral

Subprime mortgages differ radically from prime conventional loans. On average, subprime pools have lower credit scores, smaller balances, and higher LTVs than conventional pools. In addition, subprime origination is dominated by hybrid ARMs, cash out refinances, and loans with prepayment penalties. However, loan characteristics can vary widely among different subprime lenders and product types. In the following sections, we construct statistics for subprime loan characteristics using securitized 2006 vintage subprime collateral. We also further stratify characteristics by issuer, specifically highlighting seven large originator shelves. In each section, we display results for ARM and fixed-rate collateral types separately to underline differences between the loan products. All statistics are weighted by original balance.

Product Type

ARM mortgages represent the vast majority of subprime origination and comprise 40% of origination volume as shown in Figure 6. IO ARM products represent 15% of the overall universe, but make up a substantial 30% of First Franklin production. Fixed rate mortgages represent 16% of our sample. The most noteworthy recent development is the growth of the 40/30 ARM product as it constitutes 19% of 2006 originations. Subsequent sections combine ARM, IO ARM, and 40/30 ARM loans into a single “ARM” designation, and fixed and IO fixed loans into a single “Fixed” designation for brevity.

Virtually all subprime ARM loans are hybrid ARMs with a 2-year initial fixed-rate period and a periodic reset (indexed to 6-month LIBOR) every six months thereafter. These so-called “2/28” hybrids represent 83% of ARM production as shown in Figure 7. Initial fixed terms of 36 months are also frequently issued (called “3/27” hybrids), but subprime ARMs with other fixed terms are rare.

A significant percentage of subprime mortgages are first liens, with second lien products comprising 11% of production as in Figure 8. Almost all of the origination in second liens takes place through fixed-rate loans as they constitute 27% of the fixed-rate origination volume. Much of the second lien origination comes in the form of “piggyback” loans. Piggyback loans are second liens issued simultaneously with first liens with the purpose of lowering the LTV of the first mortgage. Within their fixed rate production, 55% of Fremont and 41% of Long Beach are second liens.

Figure 6: Product Type Distribution for Securitized 2006 Subprime Collateral

Product	Universe	Ameriquest	Countrywide	Fremont	First	Long	Option	
					Franklin	Beach	One	RFC
ARM	40%	60%	33%	45%	35%	62%	37%	44%
Fixed	16%	12%	29%	17%	12%	17%	12%	17%
IO ARM	15%	16%	16%	8%	30%	8%	9%	15%
IO Fixed	2%	1%	5%	0%	2%	0%	0%	3%
40/30 ARM	19%	10%	1%	27%	12%	12%	38%	16%
Balloon	8%	1%	17%	4%	9%	1%	3%	5%
Others	0%	0%	0%	0%	0%	0%	0%	0%

Source: Loan Performance, Banc of America Securities

Figure 7: First Rate Reset Distribution for Securitized 2006 Subprime Hybrid ARMs

Product	Rate								
	Reset	Universe	Ameriquet	Countrywide	Fremont	First Franklin	Long Beach	Option One	RFC
ARM	24	83%	73%	89%	96%	75%	77%	91%	66%
	36	11%	27%	6%	4%	22%	8%	4%	13%
	60	3%	0%	2%	1%	3%	15%	5%	4%
	Other	3%	0%	3%	0%	1%	0%	0%	17%

Source: Loan Performance, Banc of America Securities

Figure 8: Lien Position Distribution for Securitized 2006 Subprime Collateral

Product	Lien								
	Position	Universe	Ameriquet	Countrywide	Fremont	First Franklin	Long Beach	Option One	RFC
All	1	89%	99%	74%	91%	94%	93%	96%	88%
	2	11%	1%	26%	9%	6%	7%	4%	12%
ARM	1	100%	100%	100%	100%	100%	100%	100%	96%
	2	0%	0%	0%	0%	0%	0%	0%	4%
Fixed	1	73%	93%	72%	45%	100%	59%	66%	73%
	2	27%	7%	28%	55%	0%	41%	34%	27%

Source: Loan Performance, Banc of America Securities

Credit Scores

Credit scores are much lower for subprime mortgages than for prime conforming loans. The average FICO score in our sample is 630, about 100 points less than the average FICO score in conventional pools. Fixed-rate loans typically have higher scores than ARM loans, averaging 640 versus 624, respectively. Average FICO scores vary widely by issuer, ranging from 608 for Option One collateral to 650 for First Franklin collateral.

FICO scores also show a wide dispersion around the mean. Notice that 19% of origination has a FICO score of 680 or more, a score that is usually equated with prime quality collateral. In addition, 12% of origination has a FICO score under 560, a score normally associated with “deep subprime” credit. Some issuers exhibit greater dispersion in scores than others. Option One has a relatively tight distribution with only 8% of origination with FICO scores over 680 and only 16% of origination with FICO scores under 560. In contrast, Countrywide collateral is widely dispersed among low and higher FICO borrowers, with 14% possessing FICO scores under 560 and 23% with FICO scores over 680.

The higher FICO scores observed in fixed-rate collateral are partially the result of the emergence of “alt B” lending. The “alt B” subsector consists of originations by subprime lenders to borrowers with credit scores approaching those of prime quality “alt-A” borrowers. “Alt-B” collateral pools can have average FICO scores exceeding 670.

Figure 9: FICO Distributions for Securitized 2006 Subprime Collateral

Product	FICO Range	Universe	Ameriquest	Countrywide	Fremont	First Franklin	Long Beach	Option One	RFC
All	<560	12%	20%	14%	16%	2%	10%	16%	8%
	>=680	19%	14%	23%	18%	25%	20%	8%	25%
	560-599	17%	18%	20%	16%	11%	11%	27%	15%
	600-639	28%	29%	24%	26%	28%	33%	31%	29%
	640-679	23%	19%	20%	24%	34%	26%	18%	23%
	Mean	630	615	632	626	650	635	608	641
ARM	<560	14%	21%	18%	19%	2%	11%	17%	9%
	>=680	16%	13%	10%	16%	25%	19%	8%	23%
	560-599	19%	18%	25%	18%	10%	12%	27%	16%
	600-639	29%	29%	27%	26%	29%	32%	30%	30%
	640-679	22%	19%	19%	22%	35%	26%	18%	22%
	Mean	624	614	610	619	651	633	606	637
Fixed	<560	9%	17%	13%	4%	6%	6%	7%	5%
	>=680	25%	18%	25%	26%	24%	22%	9%	36%
	560-599	16%	17%	19%	8%	18%	9%	28%	11%
	600-639	27%	29%	23%	29%	25%	35%	34%	22%
	640-679	24%	20%	19%	33%	27%	29%	22%	26%
	Mean	640	625	636	652	644	644	617	657

Source: Loan Performance, Banc of America Securities

Documentation

“Full Documentation” borrowers constitute 54% of origination, with documentation levels being higher for fixed-rate versus ARM collateral as in Figure 10. However, documentation level varies greatly by issuer. For example, 52% of Long Beach origination is “Low Documentation” or “No Documentation,” versus 34% for First Franklin origination.

Figure 10: Documentation Distributions for Securitized 2006 Subprime Collateral

Product	Doc Level	Universe	Ameriquest	Countrywide	Fremont	First Franklin	Long Beach	Option One	RFC
All	Full	54%	56%	56%	57%	66%	48%	61%	54%
	Low/No	46%	44%	44%	43%	34%	52%	39%	46%
ARM	Full	53%	53%	57%	54%	63%	45%	59%	50%
	Low/No	47%	47%	43%	46%	37%	55%	41%	50%
Fixed	Full	63%	75%	63%	65%	82%	60%	70%	66%
	Low/No	37%	25%	37%	35%	18%	40%	30%	34%

Source: Loan Performance, Banc of America Securities

LTV

On average, subprime mortgages have much higher loan-to-value¹ (LTV) ratios than prime conventional loans. The higher LTVs reflect highly levered subprime borrowers who aggressively tap home equity and are unable to make large down payments. In fact, as shown in Figure 11, 43% of all origination has original LTVs of more than 90%. Only 12% has an LTV less than 70%. ARMs typically have higher LTVs than fixed-rate mortgages, with average ratios of 85% and 79%, respectively. However, this relationship is also issuer-dependent. For example, Long Beach fixed-rate collateral has an average LTV of 86% versus 84% for ARM loans. Higher LTV second liens comprise a substantial portion of fixed-rate Long Beach production and contribute to the higher LTVs observed in this product.

¹ We use combined LTV where available

Figure 11: LTV Distributions for Securitized 2006 Subprime Collateral

Product	LTV Range	Universe	Ameriquist	Countrywide	Fremont	First Franklin	Long Beach	Option One	RFC
All	<=70	12%	13%	17%	12%	5%	10%	18%	10%
	70-80	22%	20%	19%	25%	11%	18%	24%	18%
	80-90	23%	30%	27%	21%	13%	16%	20%	24%
	90-100	43%	37%	37%	43%	70%	56%	39%	44%
	>100	0%	0%	0%	0%	0%	0%	0%	5%
	Mean	84	84	81	84	91	87	82	86
ARM	<=70	11%	12%	14%	11%	4%	8%	17%	10%
	70-80	23%	19%	19%	27%	10%	18%	25%	18%
	80-90	23%	30%	26%	22%	13%	18%	21%	26%
	90-100	43%	39%	41%	40%	73%	56%	37%	45%
	>100	0%	0%	0%	0%	0%	0%	0%	0%
	Mean	85	84	83	84	93	88	82	86
Fixed	<=70	20%	23%	25%	11%	14%	16%	21%	13%
	70-80	22%	25%	24%	15%	22%	19%	19%	18%
	80-90	21%	29%	29%	11%	17%	10%	10%	18%
	90-100	35%	23%	22%	62%	47%	55%	50%	28%
	>100	2%	0%	0%	0%	0%	0%	0%	23%
	Mean	79	76	75	86	82	84	81	87

Source: Loan Performance, Banc of America Securities

Loan Size

The average loan size in subprime collateral is significantly smaller than in prime conventional collateral. As seen in Figure 12, the average original balance is only \$160,000, about \$30,000 to \$40,000 smaller than in agency pools. However, there is wide dispersion about this mean. 38% of all outstanding loans have original balances of \$300,000 or more, while 14% of outstanding loans have balances under \$100,000. ARM loans tend to have larger balances than fixed-rate mortgages, but the difference is issuer-dependent. As can be seen in Fremont collateral, a large percentage of second liens greatly reduces the average loan size for fixed rate mortgages. Issuers that originate few or no second liens, such as Ameriquist or First Franklin, show much smaller differences between ARM and fixed-rate collateral.

Figure 12: Loan Size Distribution for Securitized 2006 Subprime Collateral

Product	Loan Size Range	Universe	Ameriquist	Countrywide	Fremont	First Franklin	Long Beach	Option One	RFC
All	<100K	14%	8%	26%	3%	14%	11%	9%	17%
	100K-200K	26%	27%	28%	19%	31%	21%	23%	30%
	200K-300K	22%	23%	21%	21%	20%	21%	18%	22%
	>=300K	38%	41%	26%	56%	35%	48%	50%	31%
	Mean	160,635	204,750	109,608	205,862	162,939	194,532	202,810	145,219
ARM	<100K	6%	6%	7%	3%	8%	5%	5%	9%
	100K-200K	26%	27%	32%	19%	30%	19%	22%	31%
	200K-300K	24%	24%	27%	21%	22%	22%	19%	24%
	>=300K	44%	43%	34%	56%	40%	53%	54%	36%
	Mean	216,694	217,062	198,370	261,051	202,225	239,909	238,900	185,513
Fixed	<100K	30%	20%	28%	37%	23%	34%	34%	38%
	100K-200K	31%	32%	28%	29%	39%	31%	30%	31%
	200K-300K	17%	19%	19%	10%	17%	13%	13%	15%
	>=300K	22%	28%	25%	24%	20%	21%	23%	17%
	Mean	106,834	146,259	101,127	98,811	133,063	103,479	94,712	95,668

Source: Loan Performance, Banc of America Securities

Occupancy

Only 7% of subprime mortgages in our sample are non-owner occupied, very consistent with prime conventional collateral as shown in Figure 13. Owner-occupied percentages are more or less the same across ARM and fixed rate products with some of the issuers like Long Beach having higher owner-occupied percentages on fixed-rate collateral.

Figure 13: Occupancy Status Distributions for Securitized 2006 Subprime Collateral

Product	Occupancy Status	Universe	Ameriquwest	Countrywide	Fremont	First Franklin	Long Beach	Option One	RFC
All	Owner	93%	91%	96%	97%	93%	91%	93%	91%
	Investor/2nd Home	7%	9%	4%	3%	7%	9%	7%	9%
ARM	Owner	93%	91%	95%	97%	93%	90%	93%	91%
	Investor/2nd Home	7%	9%	5%	3%	7%	10%	7%	9%
Fixed	Owner	94%	93%	97%	96%	96%	95%	93%	87%
	Investor/2nd Home	6%	7%	3%	4%	4%	5%	7%	13%

Source: Loan Performance, Banc of America Securities

Prepayment Penalty

As shown in Figure 14, 69% of subprime loans contain prepayment penalties. Prepayment penalties allow the borrower to reduce the interest rate of the mortgage, often by 100 bps or more. However, the penalty for prepayment can be quite severe, normally six months interest on 80% of the outstanding balance of the loan. Prepayment penalties are slightly more common in ARMs, covering 70% of ARM collateral versus 68% of fixed collateral. In hybrid ARMs, penalty terms often match the term of the fixed rate period. Hence, most penalties in ARM collateral have terms of 24 months. The most common prepayment penalty term in fixed-rate collateral is 36 months. The percentage of longer penalty terms has declined in recent years as lenders have shied away from originating such loans due to predatory lending concerns.

Figure 14: Prepayment Penalty Distributions for Securitized 2006 Subprime Collateral

Product	Has Penalty	Penalty Term	Universe	Ameriquwest	Countrywide	Fremont	First Franklin	Long Beach	Option One	RFC
All	No	N/A	31%	39%	34%	30%	39%	30%	26%	35%
	Yes	Total	69%	61%	66%	70%	61%	70%	74%	65%
		12	6%	5%	5%	5%	8%	6%	9%	9%
		24	42%	36%	33%	42%	41%	40%	53%	33%
		36	17%	19%	10%	23%	12%	24%	12%	23%
		60	3%	0%	19%	0%	0%	0%	0%	0%
		Other	0%	0%	0%	0%	0%	0%	0%	0%
ARM	No	N/A	30%	39%	27%	29%	41%	30%	26%	31%
	Yes	Total	70%	61%	73%	71%	59%	70%	74%	69%
		12	6%	5%	5%	5%	7%	6%	9%	10%
		24	53%	42%	63%	50%	46%	44%	60%	41%
		36	11%	14%	5%	16%	5%	20%	6%	17%
		60	0%	0%	0%	0%	0%	0%	0%	0%
		Other	0%	0%	0%	0%	0%	0%	0%	0%
Fixed	No	N/A	32%	39%	32%	34%	34%	30%	31%	45%
	Yes	Total	68%	61%	68%	66%	66%	70%	69%	55%
		12	7%	8%	5%	4%	12%	5%	13%	6%
		24	8%	3%	3%	1%	23%	23%	16%	6%
		36	39%	50%	17%	61%	32%	43%	40%	42%
		60	15%	0%	43%	0%	0%	0%	0%	2%
		Other	0%	0%	0%	0%	0%	0%	0%	0%

Source: Loan Performance, Banc of America Securities

Loan Purpose

As shown in Figure 15, 50% of subprime mortgages are cash out refinances, while only 5% are rate-term refinances and 45% are for home purchases. Historically, lower credit borrowers who rely on periodic equity-takeout from their properties for debt consolidation have typically done so in the subprime market, while credit-impaired purchase or rate-term refinance borrowers often had more attractive options in the mortgage market through government-backed FHA loans. However, during the past 3 years, many purchase borrowers shifted to the subprime market due to the simpler application process (relative to FHA's) and the availability of "affordability" mortgage products like IO loans and 40/30 loans. This shift in the preference of purchase borrowers and a higher level of cash-out refinancing activity due to robust home price appreciation and low interest rates were the two of the key drivers behind explosive growth in the subprime market.

Figure 15: Loan Purpose Distribution for Securitized 2006 Subprime Collateral

Product	Loan Purpose	First Long Option							
		Universe	Ameriquist	Countrywide	Fremont	Franklin	Beach	One	RFC
All	Purchase	45%	39%	35%	70%	45%	57%	35%	43%
	Refi (Cash Out)	50%	57%	58%	27%	54%	38%	58%	51%
	Refi (No Cash Out)	5%	4%	7%	3%	1%	5%	8%	7%
ARM	Purchase	47%	42%	46%	73%	45%	59%	34%	46%
	Refi (Cash Out)	48%	55%	51%	25%	54%	37%	59%	48%
	Refi (No Cash Out)	4%	3%	3%	2%	1%	4%	7%	6%
Fixed	Purchase	30%	20%	13%	46%	51%	50%	43%	29%
	Refi (Cash Out)	62%	71%	78%	45%	47%	44%	46%	62%
	Refi (No Cash Out)	8%	9%	10%	9%	2%	7%	10%	9%

Source: Loan Performance, Banc of America Securities

Geographic Distribution

Subprime origination is concentrated in California, comprising 27% of origination volume as shown Figure 16. This compares with an average of 13% in prime conventional pools. The degree of California concentration varies over issuers. For example, 39% of Long Beach originations are California loans, but only 15% of RFC originations are from California. Geographically diverse pools are less susceptible to regional economic shocks that can affect the credit performance of pools.

Figure 16: California Concentration For Securitized 2006 Subprime Collateral

Product	First Long Option							
	Universe	Ameriquist	Countrywide	Fremont	Franklin	Beach	One	RFC
All	27%	27%	24%	23%	25%	39%	29%	15%
ARM	28%	29%	24%	25%	25%	40%	30%	18%
Fixed	21%	17%	25%	11%	26%	34%	18%	6%

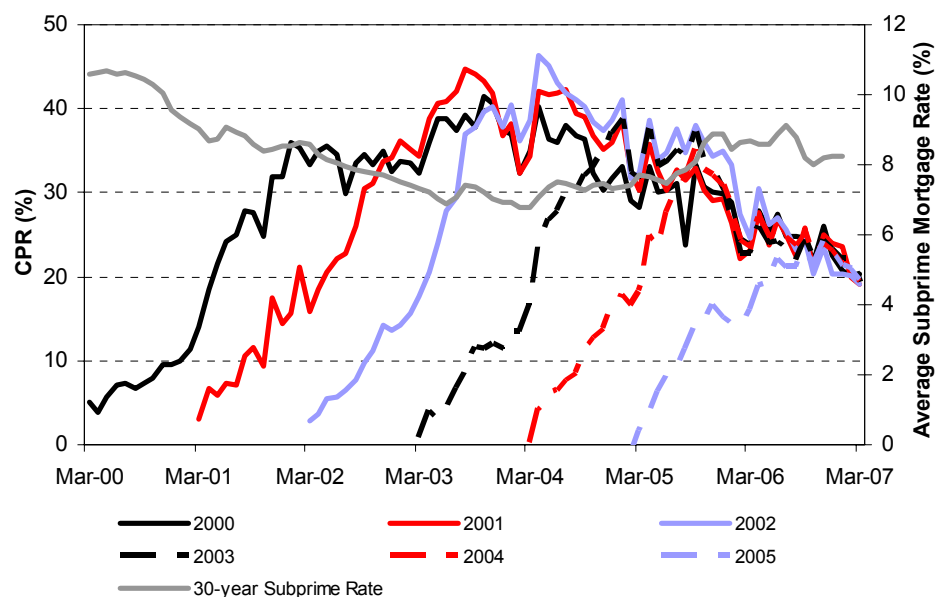
Source: Loan Performance, Banc of America Securities

IV. Prepayment Characteristics of Subprime Pools

It is well known that prepayment rates on subprime collateral are not as sensitive to changes in interest rates as prepayment rates on prime collateral. This relative insensitivity to interest rates results from a combination of many factors. Chief among these is the weaker credit profile of subprime borrowers that inevitably results in fewer available refinancing options. Weaker credit also increases the probability of default and involuntary prepayments help to prop up baseline speeds in a market sell-off. Second, subprime borrowers simply do not exercise their option to prepay as efficiently as conventional borrowers. Even when subprime market rates have dropped, these less savvy borrowers often do not take advantage of the cost benefits of refinancing. In addition, subprime borrowers tend to be more highly leveraged and the equity build-up in their homes can thus be a vital source of liquidity. Cash out refinancing, even when it results in a higher interest rate, is a necessary source of funds for many borrowers with subprime credit. Finally, subprime borrowers, who can pay over 500 bps above conventional rates, can improve their credit and qualify for a superior mortgage rate. By making consistent and timely mortgage payments, a subprime borrower may be able to obtain a better interest rate even if prevailing market rates have substantially increased. “Credit curing”—improving one’s credit grade to qualify for a cheaper mortgage rate—has a huge impact on subprime prepayments rates in the first few years of collateral seasoning.

All of this is not to say that that subprime prepayments are not all impacted by changes in mortgage rates. A market sell-off will result in prepayment speeds slowing down while a market rally will increase refinancing activity. This effect is most pronounced in fixed-rate collateral. Figure 17 shows vintage level, fixed-rate subprime prepayment speeds versus the prevailing 30-year subprime mortgage rate over time. Notice that as rates rallied in 2003, fixed-rate prepayments accelerated, reaching average speeds of 40% CPR in 2003. When rates sold off in 2006, fixed-rate subprime prepayment speeds for the 2003, 2004, and 2005 vintages slowed to the mid- to high teens. These responses to refinancing incentive are quite moderate compared to conventional pools. Agency prepayment speeds can slow to less than 10% CPR in a market sell off and surge to over 70% CPR in refinance wave. The relatively benign convexity profile of subprime pools results in a much smaller option cost.

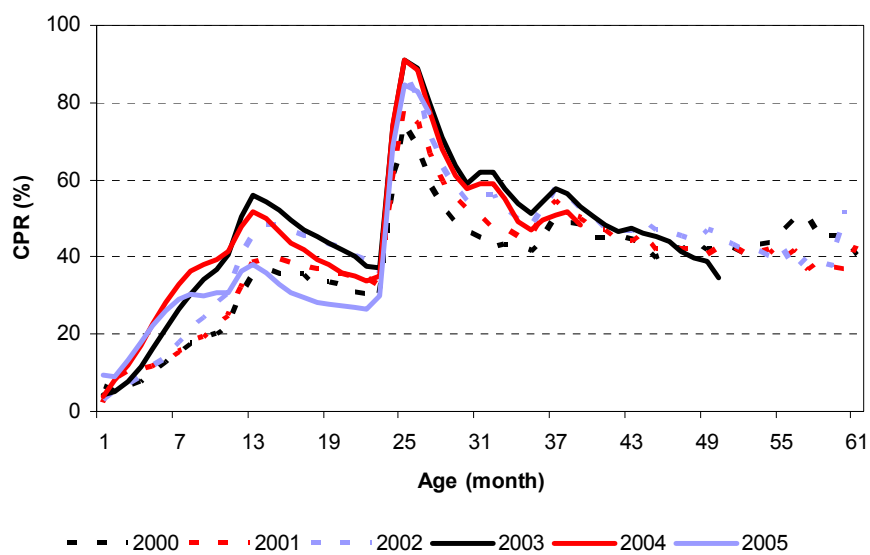
Figure 17: Evolution of Prepayment Speeds for Various Subprime Fixed Rate Vintages and the Prevailing Subprime Mortgage Rate



Source: Loan Performance, Banc of America Securities

Hybrid ARM subprime prepayments are less sensitive to long-term interest rates than prepayments on fixed rate subprime loans and are more impacted by structural features and collateral seasoning. Specifically, rate resets and any resulting payment shocks coupled with prepayment penalty expiration are far more important in determining ultimate prepayment speeds. Figure 18 shows vintage level 2/28 ARM subprime prepayment speeds by loan age. Clearly, subprime hybrid ARMs show the greatest interest rate sensitivity during the fixed rate term. Notice the relatively slow speeds of 2005 collateral prior to month 25 corresponding to the rising rate environment of 2006. However, after month 24, speeds greatly accelerate. The surge in prepayment speeds across all vintages is the result of rate resets in 2/28s and the expiration of 24-month prepayment penalties. A similar but more muted surge is evident in month 37 due to the expiration of 3-year penalty in some 2/28 loans. After month 25, notice how speeds vary in a much smaller band across vintages, in contrast to before the reset, even though the interest rate environments were radically different. However, a complete dissection of subprime prepayment patterns is a complex topic involving many nuances outside the scope of this article.

Figure 18: 2/28 ARM Subprime Prepayment Speeds by Loan Age



Source: Loan Performance, Banc of America Securities

V. Credit Performance

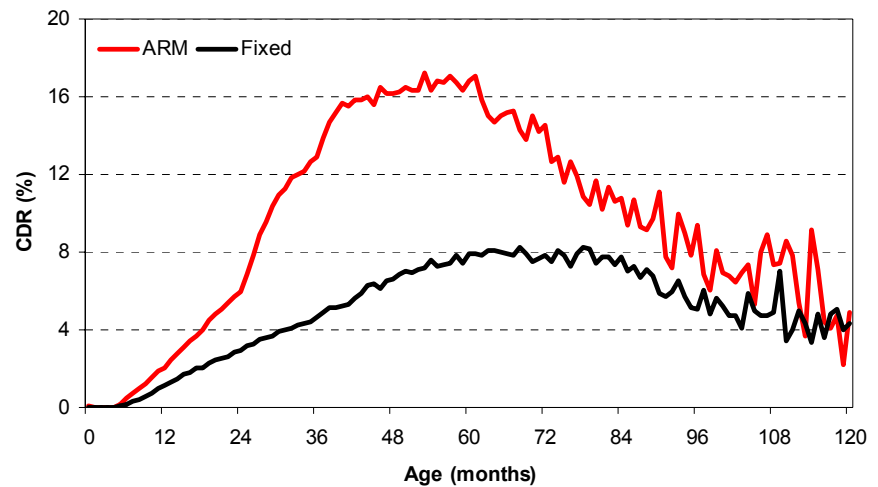
Subprime collateral pools experience elevated delinquency, default, and loss rates compared to prime quality mortgages. Figure 19 shows delinquency rates for securitized non-conforming subprime, alt-A and jumbo collateral for the 2000 through 2006 vintages as of April 2007. Notice the elevated delinquency rates in subprime collateral across all vintages. Delinquency rates for loans 60 days or more delinquent (including foreclosure and REO) are on average four times higher in subprime collateral than in alt-A, and 25 times higher than in jumbo collateral. However, in subprime collateral, delinquencies are not always a precursor to default. Subprime mortgagors will often use their loans as a cheap source of liquidity and borrow against it by skipping payments. The resulting late fees, accrued interest, and damaged credit may be the cheapest source of funds for the subprime mortgagor. “Rolling delinquencies,” where the borrower is cash flowing but remains contractually delinquent, are common.

The higher delinquency rates on subprime loans manifest themselves in more frequent defaults. Figure 20 shows historical conditional default rates for subprime ARM and fixed-rate mortgages originated since 1997. In fixed-rate collateral, default rates approach 8% CDR after a 60-month seasoning ramp. In contrast, ARM collateral default rates ramp up more quickly, exceeding 16% CDR at 48 months of seasoning. The much higher conditional default rates of ARM collateral are largely the result of faster prepayments. High prepayment rates after first rate resets and penalty expiration results in severe adverse borrower selection. When analyzed on a cumulative loss basis, fixed-rates actually underperform ARMs. Figure 21 shows cumulative losses on securitized subprime loans originated since 1997. Notice how cumulative losses are higher on fixed-rates, reaching 370bps of original balance compared to 330bps for ARM collateral after 120 months of seasoning.

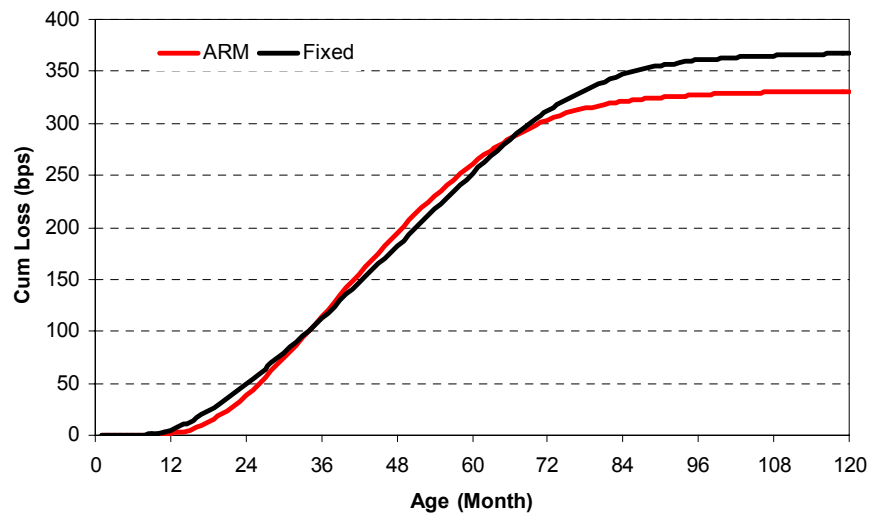
Figure 19: Delinquency Rates on Various Non-conforming Mortgage Products: 2000-2006 Vintages

Sector	Status	2000	2001	2002	2003	2004	2005	2006
Subprime	Current	72%	71%	75%	82%	81%	85%	89%
	31-60 days deliq	6%	6%	5%	3%	4%	4%	3%
	61-90 days deliq	3%	3%	2%	2%	2%	2%	2%
	91+ days deliq	9%	10%	8%	4%	4%	3%	2%
	Foreclosure	6%	7%	6%	5%	6%	5%	3%
	REO	4%	4%	4%	3%	4%	2%	1%
Alt-A	Current	88%	89%	93%	96%	94%	94%	94%
	31-60 days deliq	5%	4%	3%	2%	2%	3%	3%
	61-90 days deliq	2%	1%	1%	1%	1%	1%	1%
	91+ days deliq	2%	2%	1%	1%	1%	1%	1%
	Foreclosure	2%	2%	1%	1%	1%	1%	1%
	REO	1%	1%	1%	0%	1%	1%	0%
Jumbo	Current	95%	97%	98%	99%	98%	99%	99%
	31-60 days deliq	2%	2%	1%	1%	1%	1%	1%
	61-90 days deliq	1%	0%	0%	0%	0%	0%	0%
	91+ days deliq	1%	1%	0%	0%	0%	0%	0%
	Foreclosure	1%	0%	0%	0%	0%	0%	0%
	REO	0%	0%	0%	0%	0%	0%	0%

Source: Loan Performance, Banc of America Securities

Figure 20: Historical Subprime Default Rates by Loan Age

Source: Loan Performance, Banc of America Securities

Figure 21: Cumulative Losses by Loan Age

Source: Loan Performance, Banc of America Securities

However, there are no standard default or loss curves that can capture loss expectation across all subprime pools. At the risk of stating the obvious, lower credit borrowers will have higher default rates than higher credit borrowers. In addition, it is less likely that a borrower will default on a mortgage when sufficient equity build-up has occurred in the property. Therefore, the wide dispersion in credit scores and LTVs among subprime pools implies large variations in loss and default expectations not only for the same product but also across different products.

VI. Subprime Mortgage Servicing

Mortgage servicers can have a substantial impact on the credit performance of a collateral pool. Servicing capabilities become even more important where subprime pools are concerned because of the elevated credit risk profile of subprime borrowers. As a matter of fact, subprime servicers often follow special strategies that are not ordinarily utilized by prime servicers and will face intense scrutiny this year as mortgage credit performance continues to deteriorate in the face of a housing slowdown. In this section, we list the largest players in the subprime servicing market, illustrate the importance of servicing, detail techniques used in subprime mortgage servicing to minimize collateral losses, highlight key drivers of servicing costs, and finally, present ways of tracking servicer performance.

The Competitive Landscape for Subprime Servicing

Like subprime origination, subprime mortgage servicing is dominated by a relatively small number of servicers. Figure 22 shows the largest subprime mortgage servicers as of Q4 2006. The top ten servicers hold a market share of over 50%. Not surprisingly, many of the largest subprime servicers are also the largest subprime lenders. Usually, subprime mortgage originators retain servicing rights even after production is securitized or sold in the secondary market as whole loans. The only non-originator, also called “third party servicer”, in the top ten is Ocwen Financial. Third party servicers are dedicated mortgage servicers that bid for the servicing rights of third party portfolios. When servicing rights are not retained by mortgage originators but sold in whole loan sales, often to Wall Street conduits, servicing duties are often transferred to third party servicers. In addition, third party servicers often perform “specialty servicing” of difficult assets such as nonperforming loans.

Figure 22: Subprime Servicing Portfolio Sizes as of Q4 2006

Servicer	Portfolio Size (\$MM)	Market Share
Countrywide Financial	\$119,060	9.81%
Chase Home Finance	\$83,837	6.91%
Option One Mortgage Corp.	\$69,039	5.69%
Ameriquest Mortgage Corp.	\$65,000	5.35%
CitiFinancial (E)	\$64,250	5.29%
Ocwen Financial Corp.	\$52,160	4.30%
Wells Fargo Home Mtg.	\$51,321	4.23%
Homecomings (E)	\$50,849	4.19%
National City Home Loan	\$49,545	4.08%
HSBC Mortgage Services	\$49,465	4.07%

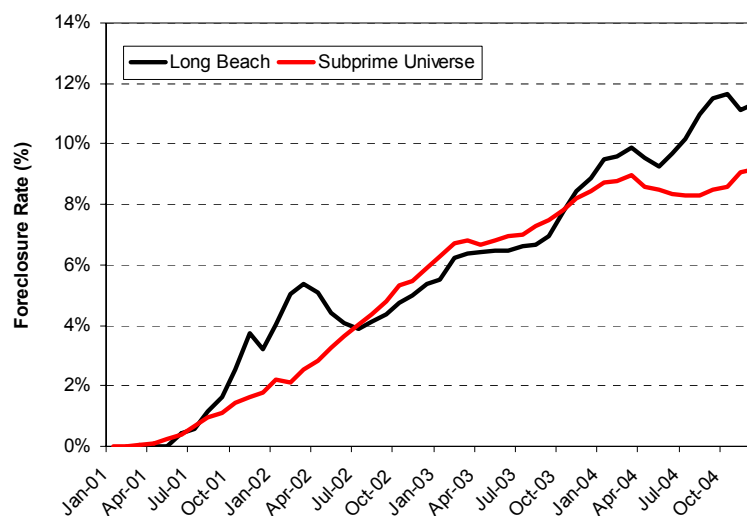
Source: National Mortgage News, Banc of America Securities, (E) are estimates

The Impact of Servicing on Collateral Performance

The dependence of subprime collateral performance on proper servicing cannot be overemphasized. Subprime borrowers require more attentive and aggressive servicing than prime borrowers. To bring this out, we walk through an example related to the servicing of Long Beach deals in 2001. Figure 23 shows foreclosure rates on 2001 vintage Long Beach collateral versus the 2001 vintage home equity universe. Notice the significant jump in foreclosure rates for Long Beach collateral from 3Q 2001 through 2Q 2002. This was a

direct result of deterioration in servicing quality. After Washington Mutual acquired Long Beach Mortgage Corporation in 4Q 1999, the Orange County based subprime servicing operation was consolidated with the company's core prime servicing operations. Many key associates were lost in the transfer leaving the subprime servicing platform deficient in expertise. Consequently, the servicing techniques employed proved ineffective on subprime borrowers and resulted in severe collateral underperformance. As a result, Washington Mutual revamped its subprime servicing in Northridge, California and installed a senior management team with extensive subprime servicing experience. The new management team emphasized employee training and hired experienced personnel from the subprime servicing industry. As a result, collateral performance improved dramatically with foreclosure rates returning to industry averages by 3Q 2002.

Figure 23: 2001 Vintage Foreclosure Rates of Long Beach and the Home Equity Universe



Source: Intex, Banc of America Securities

The Subprime Servicing Process

The two most important facets of subprime servicing are **collection management** and **default management**. Collection management involves preventing current loans from becoming delinquent and reverting early/moderate stage delinquencies to current status. Successful collection techniques involve a judicious blend of conventional wisdom and cutting edge technology. As soon as a loan is boarded on the servicing system, the customer service department does a welcome call typically within the first 30 days prior to first payment due date. This serves the twin purposes of welcoming the customer and verifying contact information. If no contact is made, the exception is referred to the skip trace unit whose sole purpose is to track down customers. The response of subprime mortgage servicers to early stage delinquencies (31-60 days) is typically more aggressive than that of prime servicers. Intelligent automated systems are used to analyze prior borrower payment behavior and prioritize collection efforts. Scoring tools such as Freddie Mac's EarlyIndicator™ enable servicers to target high-risk loans much earlier in the delinquency process. Notices of late fees are sent and followed by repeated calling campaigns until contact is made with the borrower. Call-center technology innovations

including predictive dialing systems to dramatically improve the efficacy of calling campaigns. Call-center supervisors monitor a number of key metrics to ensure efficient collections. Measures include monthly call volume, call abandon rates, average speed to answer, average talk time, and automated voice response use percentage.

If such contact attempts fail, letters demanding payment and threatening foreclosure are often sent and property inspections may be ordered. If the loan progresses to moderate stage delinquency (60 days), the collection effort is typically assumed by more experienced collectors. Contact attempts will be made on an almost daily basis and every effort is made to get the mortgage cash flowing. If all collection management efforts fail and the loan progresses to late stage delinquency (90 days), the loan is then referred to default management. Default management involves minimizing losses on seriously delinquent contracts. The primary philosophy of default management is to help borrowers retain possession of their property or sell it (based on the borrower's desire) and cure the loan default using appropriate resolution techniques. Relying too aggressively on foreclosure techniques will lead to higher collateral losses as REO liquidation often results in high loss severities. Therefore, servicers typically employ a dual strategy for late stage delinquencies: foreclosure and loss mitigation. While foreclosure is usually initiated when loans become seriously delinquent, loss mitigation techniques are simultaneously attempted in order to avoid costly REO liquidations. Once a request for foreclosure is initiated, the servicer orders a **broker price opinion (BPO)**² or an **Automated Valuation Model (AVM)**³ within the next 20 to 30 days. A good handle on the value of the property is necessary for the servicer to follow the best loss mitigation strategy. In some cases a full property appraisal may be required. In addition, the servicer will refresh the valuation of properties in foreclosure on a periodic basis, typically every six months.

Loss mitigation techniques can be categorized into two broad categories: **home retention workouts** and **voluntary title transfer workouts**. Home retention workouts include:

- **Partial reinstatement** – The borrower resumes monthly payments and enters a repayment plan that is up to 12 months in duration for past due amounts.
- **Forbearance** – Contractual suspension or reduction in monthly payments for up to 12 months followed by a repayment plan for past due amounts.
- **Loan modification** – A permanent modification of the terms of the mortgage in order to create a mortgage payment schedule that the borrower is better able to afford. Typical loan modifications include rate reduction, capitalization of missed payments, and extension of loan term or some combination thereof. Rate reduction is probably the most commonly used modification strategy. Some servicers use trial modification before making permanent changes to the contract. These trial modifications sometimes require an additional one-time upfront payment.

In the case of partial reinstatement/forbearance plans, the loan is typically reported as

² A BPO is often used in lieu of a formal appraisal to provide an estimate of the likely sale price of a property. The estimate is based on collecting a set of relevant information that would influence the property price including physical characteristics of the property (square footage, number of units, basement, lot size etc.), assessment of the property condition, comparable sales, comparable listings and neighborhood data (trends in housing supply, new construction, rental market etc.). The pricing opinion is based on an exterior “drive-by” inspection and no internal inspection is performed. The final estimate includes a range of values for the sale price based on 90-120 days marketing time including an “as-is” value and an “as-repaired” value. The opinion includes photographs of the property (and comparable listings) and a description and estimate of the necessary repairs.

³ An AVM is a computer-generated appraisal that relies on a database of comparable sales to estimate a property sales price.

delinquent to the trust. However, in the case of permanent loan modifications, the loan is usually reported as current if the borrower meets the modified terms. Identifying borrowers who have the economic ability to pay and would benefit from such workout strategies is critical. Such loan modifications are extremely common in subprime servicing portfolios. However, it is important that such loan modifications are utilized to minimize losses, not mask underperformance. Also, note that there are some constraints on when loans can be modified if these loans have been securitized as mortgage-backed securities and sold to various investors. These securities are typically issued using a tax-preferred vehicle like the Real Estate Mortgage Investment Conduit (REMIC). In order to maintain this tax-preferred status, which is absolutely critical for the investors of these securities, there are some strict restrictions that must be met on an ongoing basis. These restrictions and any additional restrictions that try to protect the interests of the investors are typically codified in a legal contract (the REMIC Pooling and Servicing Agreement (PSA)) between the trustee and the master servicer. In Appendix 1, we review legal restrictions with respect to loan modifications based on the PSAs corresponding to the 60 reference deals belonging to the three sets of ABX indices.

If home retention loss mitigation attempts fail or foreclosure is deemed inevitable due to the distressed financial situation of the borrower, voluntary title transfer workouts that involve loss of the home but avoid costly foreclosure proceedings and REO sales are attempted. Such workouts reduce high carrying costs, accrued interest, and other expenses that comprise the largest components of REO liquidations. Such techniques analyze collateral value and utilize NPV models to forecast and optimize strategies and recoveries. Voluntary title transfer workouts include:

- ◆ **Deed-in-lieu** – The borrower gives up ownership of the property in exchange for cancellation of the mortgage. “Cash-for-keys” settlements are similar but involve payment to the borrower to vacate the premises.
- ◆ **Short sale** – Also called “short payoff” or “pre-foreclosure sale”, involves negotiated sale of the property where the proceeds are less than the outstanding balance of the loan.
- ◆ **Mortgage assumption** – Assumption of the property and mortgage by a qualified applicant.

Even if loss mitigation techniques are fully utilized, some properties will inevitably be liquidated through REO sales. Therefore, timeline management in the foreclosure process is also an important component to minimizing losses. In general, the longer the timeline to liquidation, the higher the ultimate costs of foreclosure. Therefore, effective servicers are quick to initiate the foreclosure process while simultaneously pursuing loss mitigation techniques. Foreclosure timelines can vary considerably depending on the foreclosure law in each state. Broadly speaking, states are classified as judicial and non-judicial foreclosure states. Foreclosure timelines are much faster in non-judicial foreclosure states compared to judicial foreclosure states. Servicers use benchmarks such as the Freddie Mac foreclosure timelines by state to monitor the timeline management of individual loans. Once proceedings are started, the time to obtain title to the property and evict the residents should be as little as legally possible. Furthermore, expedited sales of the REO properties returning the maximum possible proceeds will have a large impact on determining ultimate loss severities.

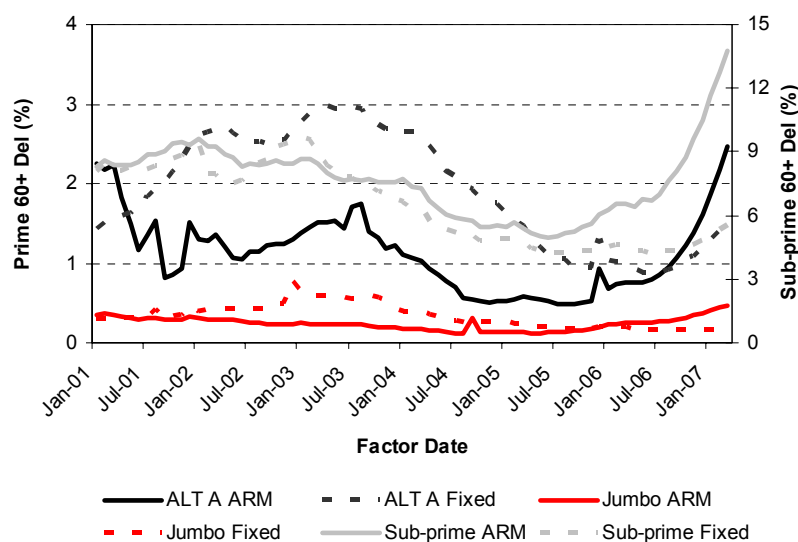
In addition to managing the foreclosure process, monitoring bankruptcies is also an aspect

of subprime servicing. In seasoned portfolios, 5%-10% of the loans could be in bankruptcy at any point in time. Typically over 60% of loans in bankruptcy are performing. Normally, a servicer will file a Proof-of-Claim within a few weeks of receiving a notification of bankruptcy. Under a Chapter 7 bankruptcy filing, the borrower gets an automatic stay initially, but cannot stop foreclosure proceedings once the lender gets court permission to start foreclosure. A Chapter 13 filing stops foreclosure indefinitely as long as the borrower continues to make monthly payments on the mortgage. In addition, the servicer will continue to monitor bankruptcies on a monthly or more frequent basis depending on the status of the loan.

The Cost of Servicing

Subprime loans are associated with much higher delinquency, foreclosure, and REO rates, which in turn result in much higher staffing requirements and costs when compared to prime loans. Figure 24 shows that 60+ delinquencies (including foreclosure and REO) are almost 20-40 times higher than jumbo loans and 6-10 times higher than alt-A loans. The increased costs are directly passed onto the subprime mortgagor in the form of higher rates as servicing fees for subprime loans averages around 50 bps while it is close to 25 bps for conventional loans.

Figure 24: Aggregate 60+ Delinquency (including Foreclosure and REO) for Various Mortgage Sectors and Products based on the OTS Method

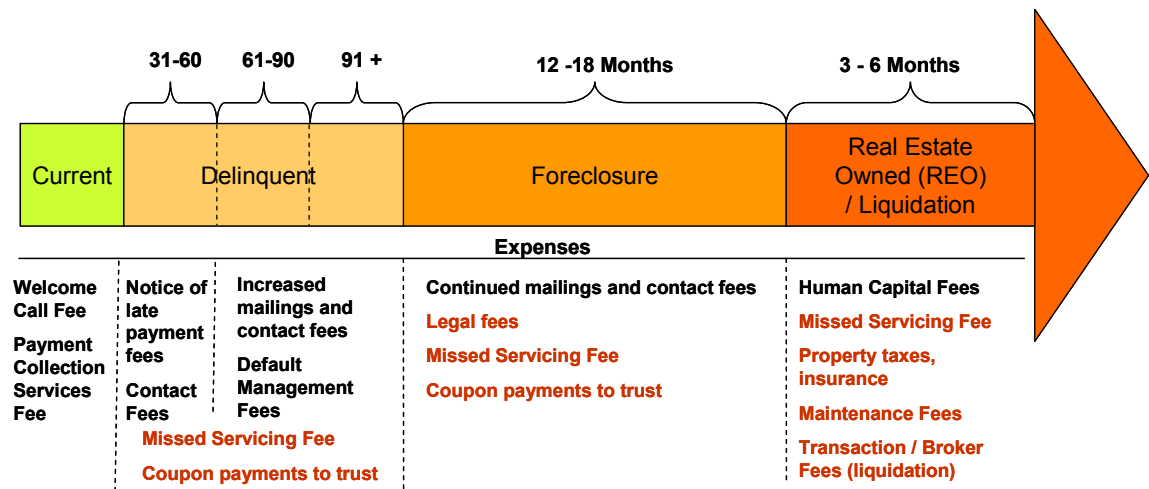


Source: Banc of America Securities

To understand the various drivers of subprime servicing costs, take a look at Figure 25. The figure shows various stages of the mortgage foreclosure process and the expenses associated with these stages. All the costs associated with mortgage servicing can be divided into two broad categories: funding costs related with reimbursable expenses and non-reimbursable operational costs. In the figure, the expenses printed in black are non-reimbursable operational costs which can be attributed to personnel, technology, and communication costs, while expenses printed in red are funding costs related to various reimbursable expenses. The reimbursable expenses are incurred at various points in time during the foreclosure process and are reimbursed without any interest when the property is

liquidated or the loan becomes current again. It should be clear from the figure that the overall level of cost associated with servicing portfolio will be directly proportional to its aggregate delinquency and foreclosure rates. Costs are typically low during the periods of good credit performance but these costs can increase very quickly during a credit downturn like the one we are facing now. Besides higher costs, servicer performance may deteriorate as servicers may not be able to fulfill their staffing needs fast enough or due to a higher level of M&A activity.

Figure 25: Expenses associated with Various Stages of Mortgage Delinquency



Source: Banc of America Securities

Monitoring Servicer Performance

As proper servicing of subprime mortgages is critical to collateral performance, investors should closely monitor servicers to ensure collateral performance falls within expectations. Three key ways in which investors monitor servicer performance are through rating agency reports and ratings, delinquency trend analysis, and roll rate analysis.

All three rating agencies publish periodic reports and ratings of mortgage servicers, including ratings specific to subprime primary servicing. Figure 26 shows subprime primary servicer ratings (as of May 2007) for the ten largest subprime servicers. The Moody’s rating scale ranges from SQ1 (strong) through SQ5 (weak). S&P rates servicers as STRONG, ABOVE AVERAGE, AVERAGE, BELOW AVERAGE, or WEAK. In addition, S&P assigns a near-term outlook of Positive, Stable, Developing, or Negative. Fitch assigns ratings of RPS1 (full approval) through RPS5 (conditional approval). Within the rating levels, Fitch differentiates the rankings by pluses (+) and minuses (-). It is interesting to see that all of the top 10 subprime servicers have solid servicer ratings from the various rating agencies.

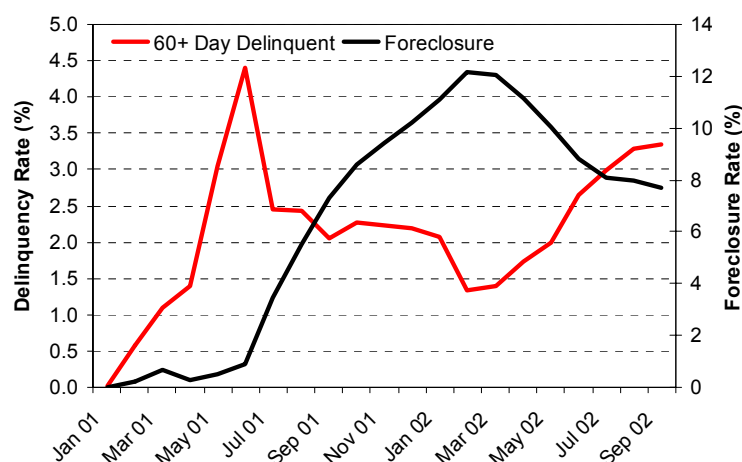
Figure 26: Current Subprime Servicing Ratings

Servicer	Moody's	S&P	Fitch
Countrywide Financial	SQ1	Strong	RPS1
Chase Home Finance	SQ1	Strong	RPS1
Option One Mortgage Corp.	SQ2+	Strong	RPS1
Ameriquest Mortgage Corp.	SQ2+	Strong	RPS3+
CitiFinancial (E)	SQ2	Above Average	RPS1-
Ocwen Financial Corp.	SQ2-	Strong	RPS2
Wells Fargo Home Mtg.	SQ1	Strong	RPS1
Homecomings (E)	SQ2+	Strong	RPS1
National City Home Loan	SQ1-	Above Average	RPS2
HSBC Mortgage Services		Strong	RPS2-

Source: Banc of America Securities

The rating agencies adjust credit enhancement requirements in subprime securitizations based on these ratings, decreasing enhancement requirements in securitizations with higher rated servicers and increasing enhancement requirements for lower rated servicers. When determining the rating, the agencies stress collection management and default management capability. However, there are many less quantifiable attributes that are also considered in the rating process. These include M&A, management experience, financial condition, staffing and training, employee turnover, operational controls and procedures, and technological sophistication. Such attributes will affect the operational and financial stability of the servicer and impact the servicer's ability to adapt to changing market conditions. For example, on April 25 2007, Moody's downgraded Option One's subprime primary servicer rating from SQ1 (strong) to SQ2+ (above average) based purely on reduction in servicing stability assessment to average from above average, in response to Option One's sale to Cerberus Capital Management, L.P. This is despite Moody's view that Option One's collection abilities are "strong" and acknowledgement that Option One had made various enhancements to its servicing platform.

As rating agencies only periodically update servicer ratings and reports, delinquency trend analysis can be used to identify servicing quality deterioration in a more timely manner. Since delinquencies are often a precursor to eventual default, unusual, steep increases in delinquency rates may reflect servicing quality deterioration that will result in higher collateral losses. We again examine the servicing difficulties experienced in Long Beach collateral beginning in 2001. Figure 27 shows 60+ day delinquency rates and foreclosure rates for Long Beach (LBMLT) 2000-1. Notice the sharp spike in delinquencies peaking in July 2001 that foreshadowed the surge in the foreclosures.

Figure 27: Long Beach 2000-1 60+ Delinquency Rates and Foreclosure Rates

Source: Intex, Banc of America Securities

Deeper insights into servicer performance and strategies can be obtained through roll rate analysis. Roll rates indicate the direction and velocity of loan status migration in a collateral pool. Specifically, roll rates show the loan status distribution of a group loans that shared a common loan status at a prior date. From these rates we can infer the percentage of loans that moved from a less serious delinquency status to a more serious delinquency status at a later date. Likewise, we can deduce the percentage of loans that returned to a current status from late stage delinquency or foreclosure. For example, Figure 28 shows average monthly roll rates for Long Beach 2000-1 collateral from 2001 to 2002. We can infer that on average, 4% of current loans migrated to 31-60 day delinquency status each month. Conversely, 35% of 31 to 60 day delinquent loans returned to current status. Roll rates can be used to monitor the effectiveness of collection management and default management. Lower roll rates from current status to delinquent status would show superior collection management. Likewise, higher roll rates from foreclosure or 91+ day delinquency statuses to current status would indicate application of loss mitigation techniques.

Figure 28: Average Monthly Roll Rates for Long Beach 2000-1 from 2001 through 2002

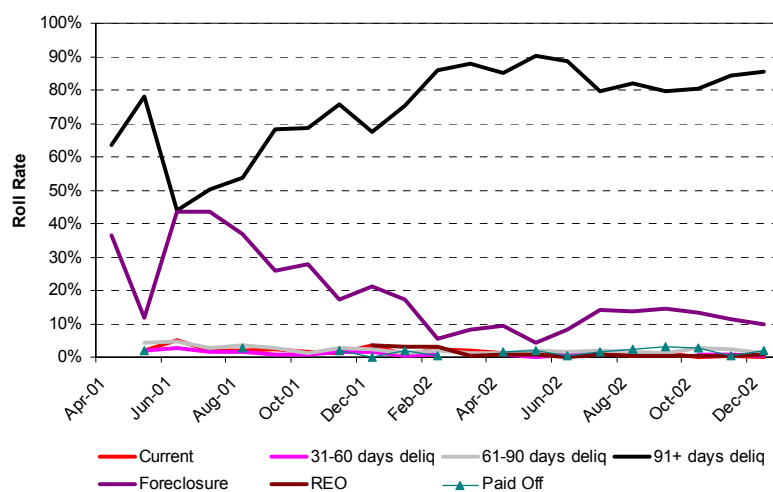
From Status	To Status						
	Current	31-60 days deliq	61-90 days deliq	91+ days deliq	Foreclosure	REO	Paid Off
Current	93%	4%	0%	0%	0%	0%	2%
31-60 days deliq	35%	27%	21%	1%	12%	0%	3%
61-90 days deliq	10%	12%	19%	40%	16%	0%	2%
91+ days deliq	1%	1%	2%	79%	15%	1%	1%
Foreclosure	3%	1%	0%	6%	83%	4%	3%
REO	0%	0%	0%	1%	1%	90%	8%

Source: LoanPerformance, Banc of America Securities

Monitoring changes in roll rates over time can be useful in analyzing the evolution of servicing procedures and capability. For example, an increase in monthly roll rates from current status to delinquent status could indicate deterioration in servicing effectiveness. In addition, changes in transitions of seriously delinquent loans could indicate modification of

loss mitigation techniques. Figure 29 shows monthly roll rates from 91+ days delinquent state to various other states for Long Beach 2000-1 from April 2001 to December 2002. Notice how the roll rates from 91+ days to foreclosure change over time. In June 2001, over 40% of 91+ day delinquent loans migrated to foreclosure status in July. However, transitions to foreclosure fell to under 10% in February 2002. Such dramatic changes in roll rates on the same portfolio reflect fundamental shifts in servicing procedures.

Figure 29: Changes in Roll Rates for Loan Beach 2000-1 from 91+ Delinquent State to Various Other States Over Time



Source: Loan Performance, Banc of America Securities

VII. The Regulation of Subprime Lending

There are no explicit rules governing subprime lending because, as yet, there is no universally accepted definition of who a subprime borrower is. Instead, regulators have targeted a number of questionable and abusive practices which have been placed under the general umbrella of “predatory lending.” The relationship between predatory lending and subprime lending stems from the belief that predatory practices are especially endemic to the subprime industry. The rationale is that borrowers with limited or damaged credit histories are especially vulnerable to being only granted access to mortgage credit at potentially extortionary rates. The subprime lending industry counters that the abuses associated with predatory lending are isolated incidents and that, in fact, subprime lending results in the provision of credit to borrowers who would typically be excluded from the mortgage market. Thus, one argument frequently made against attempts to legislate subprime lending is that this legislation could potentially result in the denial of mortgage credit to the very borrowers such laws try to protect. The debate over predatory lending and anti-predatory lending legislation is still an evolving one and has serious implications for the future of subprime lending. One particular issue of concern to secondary market participants is that some of the initial attempts at crafting anti-predatory legislation at the state-level exposed RMBS investors to significant liability from borrowers who were victimized by predatory loans. This is the notion of “assignee liability” and is one of the topics we discuss in this short introduction to the regulation of subprime lending. Since we are not legal experts, our discussion is at a high-level with the goal being to provide the reader with a conceptual understanding of the current regulatory regime without drowning in a sea of legalese.

What Is Predatory Lending?⁴

As yet, there is no precise definition of predatory lending although, as the term suggests, it encompasses practices that exploit consumers that are going through the mortgage origination process.⁵ More explicitly, predatory lending is associated with a broker or a lender taking advantage of a borrower through deception, fraud, aggressive sales tactics, or taking unfair advantage of the borrower’s lack of understanding of loan terms. Some of the practices termed as predatory lending can be cataloged as follows:

- ▶ **Flipping:** Repeatedly refinancing a borrower to collect origination fees or prepayment penalties when refinancing is not in the borrower’s best interest. Typically, each successive refinancing reduces (“strips”) the borrower’s equity.
- ▶ **Excessive fees and “packing”:** Charging fees in excess of economic justification or “packing” fees into the loan without the borrower’s knowledge or understanding.
- ▶ **Lending without regard of ability to repay:** Providing mortgages to borrowers with sizeable amounts of home equity but without the capacity to repay their mortgage in order to collect origination fees and/or foreclose on the property.

⁴ For further details see: “Consumer Protection: Federal and State Agencies Face Challenges in Combating Predatory Lending”, GAO-04-280, January 2004; “Curbing Predatory Home Mortgage Lending”, June 20th, 2000, Departments of the Treasury and Housing and Urban Development; and “A Tale of Three Markets: The Law and Economics of Predatory Lending,” Texas Law Review, Vol. 80, No. 6, May 2005.

⁵ Actually, the concept of predatory lending can be applied to any kind of consumer lending including payday lending, automobile loans, credit card lending, etc.

- ▶ **Fraud and abuse:** Outright fraud (inflated appraisals, altered settlement and application documents) or preying on vulnerable with misleading or overly aggressive sales tactics.
- ▶ **Excessive interest rates:** Charging a mortgage rate that is higher than the appropriate risk premium that the market would charge a borrower with that credit profile. Also, “steering” borrowers towards high-cost loans.
- ▶ **Single-premium credit insurance:** Credit life insurance pays off a borrower’s loan in the event of death. Borrowers can pay for coverage on a single-premium or monthly basis. Single-premium coverage is viewed as predatory for the following reasons:
 - It appears as a one-time charge on closing documents rather than each month which can make it hard for a borrower to detect and understand this charge.
 - It is expensive and typically has to be rolled into the loan which means that the borrower ends up paying interest on the premiums.
- ▶ **Prepayment Penalties:** Prepayment penalties are not regarded as being inherently abusive but can be used to lock borrowers into high-cost loans, particularly when the term of a prepayment penalty extends beyond the first reset date of a subprime ARM loan.
- ▶ **Balloon payments:** Low monthly payments without disclosure of a balloon payment.
- ▶ **Mandatory Arbitration Clauses:** Homeowners can typically litigate if they believe that their rights have been violated by the lender. Borrowers can be denied the option to litigate through a BMA (“binding mandatory arbitration”) which means that any dispute that arises in connection with the loan must be resolved through a private legal system.

Predatory practices can also occur in the context of loan servicing. For example, a servicer may charge a borrower improper late fees, require unjustified homeowner’s insurance, or not properly credit payments.

The Federal Regulation of Predatory Lending

Until the introduction of the Homeownership and Equity Protection Act (HOEPA) in 1994, the Federal government had no specific laws in place to combat the predatory lending practices outlined above. Instead, consumer protection was enforced through an umbrella of laws such as the Truth in Lending Act (TILA), Real Estate Settlement Procedures Act (RESPA), and the Federal Trade Commission (FTC) Act.⁶ Both TILA and RESPA can be broadly described as an attempt to make the costs associated with the mortgage origination process more transparent by requiring the originator of a loan provide detailed disclosures on these costs to the borrower. TILA focuses on the financing costs associated with taking out a mortgage while RESPA is concerned with closing costs (“settlement costs”).

HOEPA is an amendment to TILA that attempted to sharpen the focus of the existing regulatory framework by specifically addressing predatory lending practices. Before we get down to discussing some of the specifics of HOEPA, we first outline a simple framework for understanding HOEPA and other anti-predatory lending laws. Basically, all of these laws have three aspects to them:

⁶ Other Federal laws such as the Fair Housing Act, Equal Credit Opportunity Act (ECOA), Home Mortgage Disclosure Act (HMDA) and Community Reinvestment Act (CRA) are also used to combat predatory lending but to a lesser extent.

1. **Coverage:** The types of residential loans that are covered by the law;
2. **Restrictions:** For the loans that are covered by the law, the types of restrictions that apply; and,
3. **Enforcement:** The liability that purchasers or other assignees of mortgages face as a result of violations of the law by originators or mortgage brokers.

We now take a close look at HOEPA relative to this framework. In terms of coverage, HOEPA is targeted at “high cost” loans. These are defined as loans where the (i) the APR exceeds the Treasury rate of comparable maturity by more than a specified amount (the “triggers” are currently 8% for first mortgages and 10% for seconds), or (ii) points and fees on the loan exceed \$547 or 8% of the loan amount, whichever is greater. (The \$547 figure is for 2007 and is adjusted annually by the Federal Reserve Board based on inflation.) Credit insurance premiums are counted as fees in this context. Also, HOEPA *only* applies to closed-end refinance transactions on owner-occupied primary residences. This means that a HELOC or a purchase loan would not fall under HOEPA.

For loans that fall under HOEPA, the law imposes significant restrictions. Some of the salient ones include a prohibition against most prepayment penalties, a prohibition against balloon payments for loans with terms less than 5 years and a prohibition against loans with negative amortization features. HOEPA loans also require additional disclosures including a written notice that the loan need not be completed, the borrower could lose the residence if they fail to make their payments, plus information on the loan terms.

Finally, from an enforcement perspective, HOEPA imposes liabilities for violations of the law on both lenders and assignees of HOEPA loans. Holders of HOEPA loans “are subject to all claims and defenses...that could be raised against the original lender.” The extent of this liability can include statutory and actual damages, court costs and attorney’s fees. In addition, in the case of a violation, the loan may be cancelled for up to three years after origination.

Even though HOEPA is the only federal law directly targeted towards predatory lending, in practice consumer protection laws like RESPA and TILA, which were briefly discussed above, have also been used to combat predatory practices. Congress provides certain federal agencies with the responsibility of writing and enforcing the regulations that implement HOEPA and these other consumer protection laws. These agencies consist of the Federal Reserve Board (the “Board”) and HUD. The Board writes Regulation Z which implements both TILA and HOEPA, while HUD writes Regulation X which implements RESPA. Other federal banking regulators include the Office of the Comptroller of the Currency (OCC), the Federal Deposit Insurance Corporation (FDIC), and the Office of Thrift Supervision (OTS). Figure 30 uses a matrix format to show the applicability of various federal laws to different predatory lending practices, along with the enforcing federal agencies. The figure drives home the point that predatory lending practices are currently regulated by a matrix of consumer protection laws and enforcement agencies.

The final piece of the federal regulatory picture consists of understanding the relationship between different participants in the mortgage industry and the federal regulators. Mortgage originators consist of depository institutions (commercial banks, thrifts and credit unions) and non-depository institutions such as securities firms and finance

companies. Figure 31 shows which of the relevant federal agencies supervises the various institutions who originate mortgage loans. One of the most important takeaways from the figure is that the nonbank subsidiaries of holding companies who have traditionally played an important role in subprime lending are not supervised by federal banking regulators. Non-federally supervised non-bank subprime lenders represented approximately 2/3rds of the subprime market at its peak over the past couple of years, although that share is likely lower now given the recent shakeout in the subprime industry. The FTC is the primary enforcer of consumer protection laws for these institutions but does not conduct supervisions.

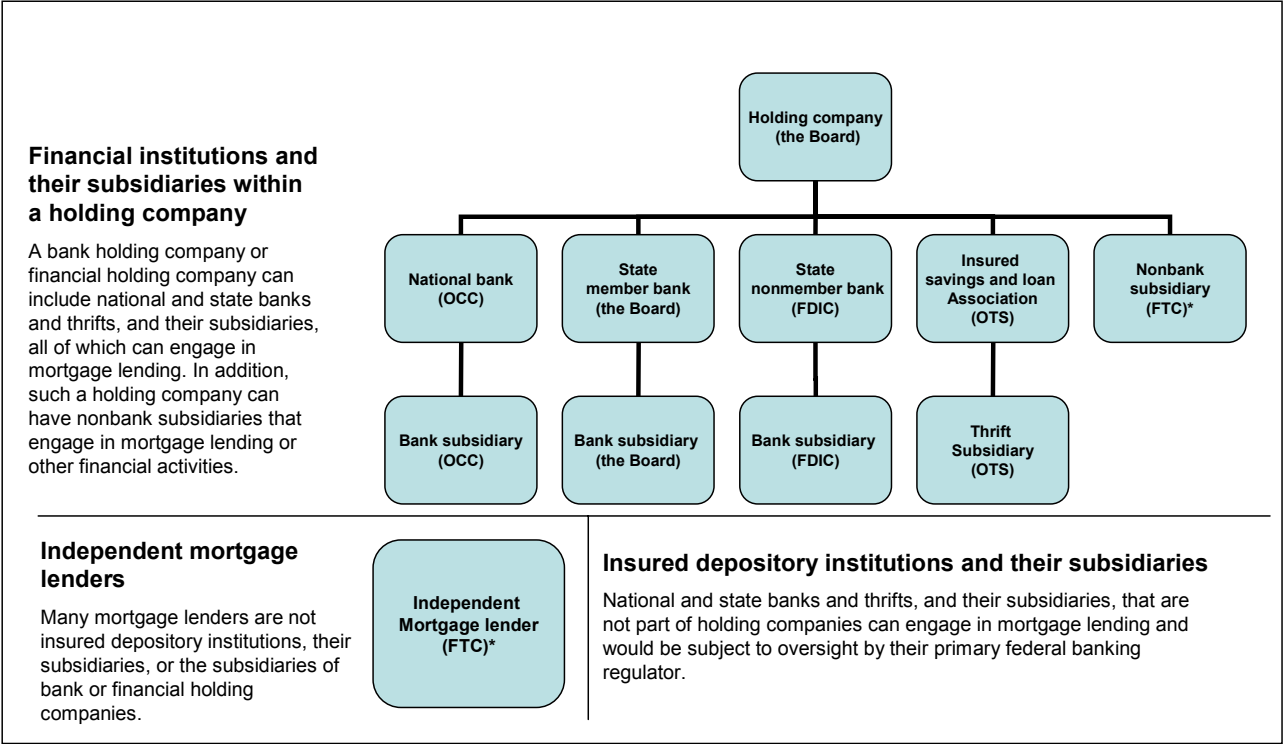
Figure 30: Federal Laws and Statues to Address Predatory Lending Practices

Predatory lending practice	GOVERNING FEDERAL STATUE					ENFORCING FEDERAL AGENCIES			
	TILA	HOEPA	RESPA	FTC Act*	Title 18 of U.S. Code	Federal banking regulators	FTC	HUD	DOJ
Failure to disclose actual loan costs	X	X	X			X	X	X	
Prohibited fees and payments			X			X		X	
Lending without regard to ability to repay		X				X	X		
Loan flipping		X				X	X		
Fraud and deception				X	X	X	X		X
Prohibited prepayment penalties		X				X	X		
Prohibited balloon payments		X				X	X		

*The FTC Act empowers the FTC to take action in situations where borrowers are misled or deceived about the terms of their loan.

Source: GAO.

Figure 31: Structure and Federal Oversight of Mortgage Lenders



*FTC is responsible for enforcing federal laws for lenders that are not depository institutions but is in not a supervisory agency and does not conduct routine examinations.

Notes: The primary federal agency for enforcement of the various federal laws used to combat abusive or predatory lending activities is shown in parentheses.

Source: GAO

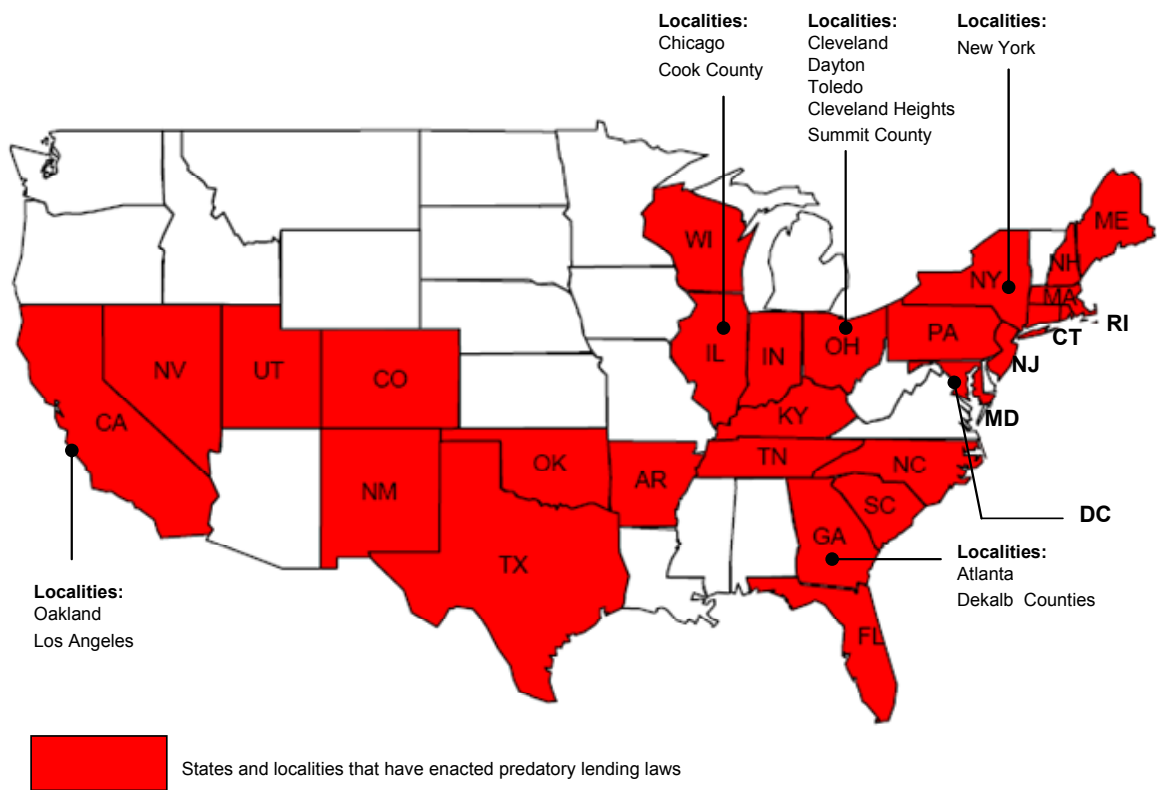
The State-level Regulation of Predatory Lending

In practice, very few subprime loans meet the APR or point/fee “triggers” of HOEPA. A recent loan level analysis of subprime data demonstrated that less than 0.5% of all refinance or home improvement loans reported for 2005 met the APR trigger.⁷ With the HOEPA triggers covering a very limited subset of subprime loans, a number of states have taken the initiative in terms of crafting more restrictive “mini-HOEPA” laws to combat predatory lending practices. The first such law was enacted by North Carolina in 1999 and took effect on July 1, 2000.⁸ In general, these state-level laws are a patchwork of different regulations that attempt to regulate the practices described in the section on predatory lending. In addition, many states have increased their regulation of brokers and lenders. Following on the heels of the North Carolina law, a number of states (and counties and cities) enacted their own legislation and Figure 32 provides a visual summary of the areas of the U.S. which currently have some form of a predatory lending law.

⁷ See “Higher-Priced Home Lending and the 2005 HMDA Data.” Robert B. Avery, Kenneth P. Brevoort, and Glenn B. Canner. September 8th 2006, Federal Reserve Bulletin.

⁸ A number of cities and counties have passed laws of their own to address predatory lending since 2000. Some of these laws have been enacted but some are still short of being enforced because of ongoing litigation.

Figure 32: States and Localities with Predatory Lending Laws



Source: Butera and Andrews (as of January 22nd, 2007).

To get a sense for the key features of these state-level laws, it is useful analyze them along the dimensions of **coverage**, **restrictions**, and **enforcement**, as in our discussion of HOEPA. For example, take a look at Figure 33, which summarizes a number of features of the North Carolina’s predatory lending law. In terms of coverage, the North Carolina law covers many more loans than HOEPA with its application to both open-end and closed-end refinance loans and purchase loans. The restrictions are also more severe with their requirement for loan counseling and their prohibition against the financing of single premium credit insurance. Finally, as we shall see in the section on assignee liability, the North Carolina law has much stronger enforcement with assignees facing a potential financial liability which is more than twice that imposed by HOEPA.

One important point to note is that other state laws can vary significantly along each of the dimensions described above or shown in the left-hand column of the table in terms of what types of loans they cover, what the triggers on the loans are etc. Indeed, one of the most problematic features of these state-level predatory lending laws is the wide variation in their requirements and restrictions. Several industry participants have criticized these laws for having the potential to significantly increase the cost of compliance and also expand the liability faced by both originators and investors for loans originated in the jurisdictions that these laws apply.

Figure 33: North Carolina Predatory Lending Law

Coverage	"High-cost" loans including HELOCs but excluding reverse mortgages, where the principal (or the borrower's initial maximum credit limit for HELOCs) is less than MIN(conforming loan limit, \$300,000.
APR Trigger	APR > T-bill + 8% for first lien; T-bill + 10% for second lien.
Points and Fees Trigger	Greater than 5% of the total loan amount for loan sizes greater than \$20,000; or greater than MIN(8% of loan amount, \$1,000) for smaller loans
Prepayment Penalties	Prohibited on all loans below \$150,000 (not only high-cost loans).
Loan Counseling	Required
Ability to repay	Total monthly debt payments do not exceed 50% of monthly gross income.
Balloon payments	Prohibited
Assignee liability	S&P indicative loss severity 275%
Financing of fees	No financing of points and fees or any charges payable to 3 rd parties. Also, no financing of prepayment fees or penalties in a refinancing by the same creditor or its affiliate.
Mandatory arbitration limiting judicial relief	No provision
Financing of credit life, disability, unemployment, or other life or health insurance premiums (except monthly premium)	Prohibits financing of single premium credit insurance (applies to all consumer home loans).
Loan flippings	Prohibited

Source: GAO, Appendix A in Ho and Pennington-Cross (2005), "*The Impact of Local Predatory Lending Laws*." Federal Reserve Bank of St. Louis Working Paper, WP 2005-049B.

Federal Preemption of State Law

There is significant disagreement between the states and federal regulators as to whether the states can bring enforcement actions against national banks or their operating subsidiaries for participating in predatory lending. Several federal regulators (specifically, OCC, OTS and NCUA) have recently determined that federal law preempts some of these state predatory lending laws. The federal preemption of state law arises from the Supremacy Clause of the U.S. Constitution which makes federal law "the supreme law of the land." Essentially, the federal regulators have argued that the state laws do not apply to institutions that come under their jurisdiction (see Figure 31) and are hence already subject to the laws outlined in Figure 31. For example, in January 2004, the OCC issued a regulation that effectively barred the application of state laws to national banks and essentially implied that national banks are immune from state predatory lending laws. Here, it should be pointed out that the state laws continue to apply to institutions that are not under the bailiwick of the federal regulator issuing the preemption order. So, for example, any promulgation issued by the OTS does not apply to independent mortgage companies, national banks and state-owned banks and thrifts. In particular, state laws continue to apply to non-bank lenders who currently play a significant role in subprime lending.

The implication of federal preemption from the viewpoint of an RMBS investor is that the rating agencies have determined that transactions that contain loans that were originated by

entities that are supervised by the federal regulators listed above do not need additional credit enhancement. For example, loans that are originated in an area that is subject to any state or local predatory law would not need additional credit enhancement if the mortgage lender is an operating subsidiary of a national bank.

The federal preemption of state law is currently a hot topic of debate with consumer advocates and other parties arguing that the states and localities are better able to police predatory lending, while one of the counterarguments from the federal side are the inefficiency and compliance costs associated with having national banks adhere to both federal law and a complex patchwork of state laws. The courts are ultimately responsible for resolving these conflicts between federal and state laws. In most recent decisions, the courts have repeatedly come out in favor of federal preemption.

Assignee Liability

We wrap up our discussion of predatory legislation by turning to the aspect of these laws that has raised the most concern among secondary market participants. The assignment of loans from originators to third parties such as loan purchasers or a securitization trust (“assignees”) is one of the cornerstones of a secondary market. The term “assignee liability” refers to provisions in the state and Federal anti-predatory lending laws that expressly hold purchasers or securitizers of loans liable for violations of the law committed by the originator and/or party to lawsuits by borrowers who were victimized by predatory loans. The goal of these provisions is obviously to discourage secondary market participants from purchasing loans that contain predatory features by exposing them to potentially heavy damages by doing so.

One significant problem with assignee liability provisions is that it is often very challenging for market participants to discern which of the loans in a pool are predatory based on their due-diligence process. In particular, note that it is very challenging for investors to discern whether borrowers who take out limited documentation SISA loans have the ability to repay these loans or not. Similarly, looking at a loan file does not permit one to determine whether a loan has been flipped or not. Another significant problem with assignee liability provisions is that they are often very broadly stated which means that investors are unable to precisely quantify the limit of their financial exposure to the loan. In other words, their exposure is “uncapped”. Needless to say, in such situations, many investors will simply refuse to purchase the loan at any price. As a matter of fact, the rating agencies have played an instrumental role in terms of raising concern about the shortcomings of some of these measures by simply refusing to rate any loans that are “covered” by existing predatory lending legislation specific to a particular jurisdiction because of their “unquantifiable” risk. For example, in January 2003, S&P announced that it would not rate any MBSs with Georgia loans because of the assignee liability provisions embedded in the state’s predatory lending law, the Georgia Fair Lending Act (GLFA). Under GLFA’s assignee liability measure, secondary market participants, brokers, and loan servicers were potentially liable for violations of the law made by the originator of the loan. The remedies available to borrowers included actual damages, cancellation of high-cost loans, attorney fees, and punitive damages. Most of the violations were civil violations, but a knowing violation would count as a criminal offense.

The response to the assignee liability measure by the rating agencies, lenders and other market participants led to the Georgia legislature rewriting GLFA to significantly reduce the scope of assignee liability. The amendments restricted the liability to “high-cost” loans and only applied when the purchaser was unable to show that it had made a good faith attempt at conducting a proper due-diligence. Furthermore, the amendments capped the amount of damage an assignee could face and prohibited assignee liability in class-action lawsuits. Once these amendments were passed, lenders were once again willing to originate in the state and the rating agencies agreed to rate MBS with Georgia loans.

The current rating agency approach to assignee liability reviews the predatory lending laws for each state and its localities in order to determine what types of loans are covered by the existing laws, how clear the laws are about how violations are determined and what the damages associated to these violations are. For example, Figure 34 shows an excerpt of a matrix that S&P produces that summarizes the key aspects of anti-predatory lending laws from the perspective of assignee liability on a state-by-state basis. Blanket exclusions of certain types of loans from structured finance transactions only apply when the state predatory lending statutes in question are not specific enough about what types of loans are covered under the law and/or if damages are uncapped. The rating agencies will also look to see if the relevant statute contains any “safe harbors”: due-diligence procedures that a purchaser or assignee can perform to avoid liability. The rating agencies also require compliance checks that verify the seller’s ability to detect a covered or predatory loan and to repurchase any loans that are in breach of anti-predatory lending laws.

The rating agencies calculate the required credit enhancement for covered loans separately. For example, Figure 34 shows that S&P associates a very high loss severity for loans that fall under the ambit of the Fed’s HOEPA law or the North Carolina law. Thus, the credit enhancement associated to these laws may make it non-economic to include them in a securitization. Furthermore, there may be additional credit enhancement required for these loans depending upon how clear and objective the applicable law is.

Modern Times

The regulation of subprime lending is an area of significant flux and controversy in the wake of a significant increase in subprime defaults and delinquencies over the past several months. A number of the concerns that consumer advocates have over predatory lending have been strengthened with one of the specific contentions being that borrowers were steered into loans that they would ultimately be unable to repay. While the coverage, restrictions and enforcement provisions of HOEPA have all been subject to contention, the list of topics that the Fed requested comment on in the HOEPA hearings that are currently underway provide some indication on the areas of focus for further rulemaking. Specifically, the Fed has asked industry participants for their comments on the following four practices:

1. Prepayment penalties: Should prepayment penalties be restricted? In particular, should prepayment penalties that extend beyond the first adjustment period on an ARM be prohibited?
2. Failure to require escrow for taxes and insurance: Should escrows for taxes and insurance be required for all subprime mortgage loans?

3. Stated-income and low-documentation lending: Should stated-income or low documentation loans be prohibited for subprime borrowers or “higher risk” loans such as loans with high loan-to-value ratios?
4. Evaluating a borrower’s ability to repay a loan: Should lenders be required to underwrite all loans based on the fully-indexed rate and fully amortizing payments? Is a loan “unaffordable” if the borrower’s debt-to-income ratio exceeds 50% at loan origination?

Thus, it appears likely that if the Fed does act, it will be to further restrict or prohibit at least one and perhaps all of the above practices in the near future. We view this as a fairly likely outcome given the current state of the housing market and the pressure being imposed by Congress on the Fed to give the consumer protection laws more teeth.

Figure 34: S&P Assignee Liability Matrix for Federal and North Carolina Anti-Predatory Lending Laws

Jurisdiction and Effective Date	Loan Categories	Servicer Violation	Subjective Standards	Mitigating Factors*	Maximum Securitization Trust Exposure	Indicative Loss Severity**	Additional Enhancement Required
Fed/HOEPA Oct 1, 1995	High Cost Loans (Section 32)	N	Y	N	Remaining balance of the loan, plus principal paid by the borrower, plus interest paid to date, plus attorney’s fees and costs	119%	Y
North Carolina	Consumer Home Loans (Refis only)	N	Y	Y	Forfeiture of interest charges plus twice the interest paid, plus attorneys’ fees and costs	275%	N
	High Cost Home Loans	Y	Y	Y	Forfeiture of interest charges plus twice the interest paid, plus attorneys’ fees and costs	275%	N

* “Y” means that S&P believes that there are mitigating factors that negate the subjectivity of the law. An “N” means either: there are no mitigating factors; or there are mitigating factors but S&P does not believe the factors successfully negate the subjective standards. Mitigating factors include: whether damages are imposed only if there is a “pattern or practice” of violating the law; whether the law requires the borrower to prove that a violation was committed “knowingly and intentionally,” whether a law provides objective standards for satisfying a repayment ability test or a net tangible benefits test, the litigation history of the law; procedural factors contained in the law, such as statutes of limitation, cure periods, rebuttable presumptions, restrictions on affirmative and defensive claims against assignees; and any other factor that S&P deems relevant.

** For the assumptions that S&P makes in its loss severity calculations see “*Anti-Predatory Lending Law Update*”, 20th October, 2006.

Source: S&P.

Appendix I: An Overview of Loan Modification Flexibility for Deals Underlying the ABX Indices

Roughly speaking, the primary constraint on loan modifications specified in the PSAs stems from the special tax and accounting status of the trust that is used to securitize mortgage loans. From a tax perspective, most trusts are established as Real Estate Mortgage Investment Conduits (REMICs) and, as such, are exempt from double taxation (only investors are taxed, not the trust itself). Now, under these same tax laws, assets may not be contributed to and/or disposed from the pool of loans securitized in a REMIC trust in order for the trust to retain this special status. In other words, the pool of loans in the trust must be static.

From an accounting perspective, the special accounting treatment of securitization trusts as qualifying special purpose entities (QSPEs) under FAS 140 exempts the depositor/seller of mortgage loans to the trust from the consolidation of the QSPE's assets and liabilities onto the seller's⁹ financial statements. However, one of the qualifying criteria under FAS 140 requires the QSPE to be on "auto pilot" and the sponsor to not have effective control over the trust.

In general, a "significant" modification of a mortgage is treated, for tax purposes, as an exchange of the old (unmodified) mortgage for a new (modified) mortgage and hence, prohibited under REMIC rules. In addition, loan modifications may also be construed as a sponsor exercising its control over the trust and therefore constitute a violation of FAS 140 rules. In short, loan modifications could result in an SPE losing its special tax and accounting status. One of the exceptions to these general prohibitions allows for loan modifications when a borrower has either defaulted or a default is "reasonably foreseeable." Thus, PSAs usually contain a broad set of allowable loan modifications for defaulted loans and a much more limited set of options for other loans.

In terms of the limited set of options for the loans that have not defaulted or where a default is not reasonably foreseeable, there are typically two types of temporary workout actions allowed by the PSAs. The first is the waiving of late payment charge or penalty interest and is allowed by every deal we examined. The second action is the extension of the payment due date. We have seen the time limit on such extensions ranging from anywhere between 120 days and 360 days, with majority of the deals allowing a 180 days extension period. The servicer, however, is required to make timely advances of interest and principal during such an extension period.

If a borrower has defaulted or a default is foreseeable (based on servicer's judgment) then extensive permanent loan modifications are allowed under most of the PSAs that we reviewed. The goal of these modifications is to maximize timely and complete recovery of principal and interest on the mortgage notes within the limits of applicable laws. Typical modifications are lower mortgage rates, extension of loan term, capitalization of missed interest payments, forgiving missed payments, or some combination of these. However, about half of the PSAs often place some limitations on loan modifications. Before going into the details of various categories of constraints that may be imposed on loan modifications, it is worth reiterating that an assessment of a foreseeable default is very subjective. It is not clear to us what legal documentation the servicer needs to maintain in

⁹ Also referred to as a "sponsor."

order to support their assessment of an imminent default. In our opinion, missed payments do provide clear evidence of a default in the foreseeable future. However, we don't think a rate reset in the near future is a reason in and of itself to characterize a loan as being in danger of imminent default.

Some of the constraints on loan modifications are applicable at the loan level while others apply on an aggregate basis. The various categories of constraints imposed on modification of loans that are in default or where default is foreseeable are as follows:

- There may be a floor on the reduced mortgage rate. The floor value may be specified on an absolute basis, as a fraction of the old mortgage rate, or based on a servicing fee.
- Restrictions on the extension of the loan maturity date (for example: limited to one year).
- Limits on the number of times a loan may be modified either on an annual basis or during the lifetime of the loan.
- The most common constraint on an aggregate basis is to restrict the total number or aggregate balance of loans that can be modified (typically 5% of the total number of loans or the total balance as of the deal cut-off date). Exceeding this cap is possible but prior approval often needs to be obtained from third-parties such as NIM insurers, P class holders, or the rating agencies.

Besides loan modifications, it may also be possible for the depositor, servicer, investors in certain classes, or the NIM insurer to buy out loans that have been delinquent for certain duration and have either defaulted, or where default is foreseeable. These loans can typically be bought out of the trust at par. However, there are some limits, either based on the aggregate loan balance or loan count (typically 5%). This action may be beneficial for bond holders as it reduces likely losses on the collateral. In Figure 3.1, we present a summary of the various loan modification constraints that were specified in PSAs corresponding to each deal.

In summary, our review of the PSAs shows that only temporary workouts are possible before default/foreseeable default. However, once the loan defaults or default is foreseeable, servicers for most deals do have extensive permanent loan modification flexibility in terms of working out a solution with borrowers in order to minimize the costs associated with handling of defaulted mortgage loans.

Figure 35: A Summary of Loan Modification Constraints on Deals in the ABX Indices

		ANY MORTGAGE LOAN	LOAN HAS DEFAULTED OR DEFAULT IS FORESEEABLE			
		Extend due date by	Rate or Term Mod Limitations	Frequency	Aggregate Limit on # or Balance (% of Original Pool)	Buyout by (If Allowed)
ABX 06-1	ACE 2005-HE7	180 days				Servicer
	AMSI 2005-R11	120 days				NIM Insurer/Servicer
	ARSI 2005-W2	180 days			5% (#)	NIM Insurer/Servicer
	BSABS 2005-HE11	125 days	Rate >= 5%			Seller
	CWL 2005-BC5	270 days			5% (#)	Seller
	FFML 2005-FF12	180 days		<= 1 last 12 mo, <= 3 life time	5% (#)	Servicer
	GSAMP 2005-HE4	180 days				Depositor
	HEAT 2005-8	180 days	No Mod Allowed			Special Servicer
	JPMAC 2005-OPT1	270 days				Servicer/NIM insurer
	LBMLT 2005-WL2	180 days			5% (#)	Holder of Class C/Servicer
	MABS 2005-NC2	180 days			5% (#)	Servicer/NIM insurer
	MLMI 2005-AR1	180 days				Servicer
	MSAC 2005-HE5	180 days				
	NCHET 2005-4	180 days				
	RAMP 2005-EFC4	No hard limit	Rate >= max(old rate/2, svc fee)	**	5% (\$)	Servicer
	RASC 2005-KS11	No hard limit	Rate >= max(old rate/2, svc fee)	**	5% (\$)	Servicer
	SABR 2005-HE1	180 days	No Mod Allowed			
ABX 06-2	SAIL 2005-HE3	120 days				NIM Insurer
	SASC 2005-WF4	120 days				NIM Insurer
	SVHE 2005-4	180 days				
	ACE 2006-NC1	180 days			5% (#)	Servicer
	ARSI 2006-W1	180 days				NIM Insurer/Servicer
	BSABS 2006-HE3	125 days	Rate >= 5%			Sponsor
	CARR 2006-NC1	180 days				
	CWL 2006-8	270 days			5% (#)	Servicer
	FFML 2006-FF4	180 days				Depositor
	GSAMP 2006-HE3	180 days				Depositor
	HEAT 2006-4	180 days	No Mod Allowed			Special Servicer
	JPMAC 2006-FRE1	270 days				Servicer/NIM insurer
	LBMLT 2006-1	180 days			5% (#)	Holder of Class C/Servicer
	MABS 2006-NC1	180 days			5% (#)	Servicer/NIM insurer
	MLMI 2006-HE1	180 days				Servicer
	MSAC 2006-WMC2	180 days	No Mod Allowed			
	MSC 2006-HE2	180 days	No Mod Allowed			
	RAMP 2006-NC2	No hard limit	Rate >= max(old rate/2, svc fee)	**	5% (\$)	Servicer
	RASC 2006-KS3	No hard limit	Rate >= max(old rate/2, svc fee)	**	5% (\$)	Holder of Class SB / Servicer
ABX 07-1	SABR 2006-OP1	360 days				
	SAIL 2006-4	120 days				NIM Insurer
	SASC 2006-WF2	120 days				NIM Insurer
	SVHE 2006-OPT5	180 days			5% (#)	Servicer
	ABFC 2006-OPT2	No hard limit	1 year ext.; rate lower by 50bps		5% (#)	NIM Insurer/Servicer
	ACE 2006-NC3	180 days				Servicer
	BSABS 2006-HE10	125 days	Rate >= 5%			Seller
	CARR 2006-NC4	180 days				Servicer
	CBASS 2006-CB6	No hard limit	1 year extension; Rate lower by 50bps			Servicer
	CMLTI 2006-WFH3	180 days				
	CWL 2006-18	270 days			5% (#)	Seller
	FFML 2006-FF13	180 days				Depositor
	FHLT 2006-3	180 days				Servicer
	GSAMP 2006-HE5	180 days				Depositor
	HEAT 2006-7	180 days	Rate >= Initial rate (ARM), No Rate Mod (FRM)			Holder of Class X/Special Servicer
	JPMAC 2006-CH2	270 days				Servicer/NIM insurer
	LBMLT 2006-6	180 days			5% (#)	Holder of Class C
	MABS 2006-NC3	180 days				Servicer/NIM insurer
	MLMI 2006-HE5	180 days			5% (#)	Servicer
	MSAC 2006-HE6	180 days	No Mod Allowed			
	RASC 2006-KS9	No hard limit	Rate >= max(old rate/2, svc fee)	**	5% (\$)	Holder of Class SB/Servicer
	SABR 2006-HE2	180 days	No Mod Allowed			
	SASC 2006-BC4	No hard limit	No Mod Allowed			NIM Insurer
	SVHE 2006-EQ1	180 days				Servicer

** Capitalization (adding to principle balance) can only happen once

Source: Banc of America Securities, SEC

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